

Sri Lanka Institute of Information Technology



Data Warehousing and Business Intelligence IT3021 B.Sc. (Hons) in Information Technology DATA SCIENCE

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Declaration

I declare that this project report, or any part of it, is not copied from any document produced by any organization, university, or other institution, or from any previous student project at SLIIT. It has also not been copied from the internet or any other sources.

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Acknowledgement

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Finally, I wish to convey my special appreciation to the **Sri Lanka Institute of Information Technology (SLIIT)** for providing the necessary resources, facilities, and a supportive learning environment to successfully complete this assignment.

Step 1: Data Source for Assignment 2

1.0 Introduction to the Data Source

The data source used for Assignment 2 is the crime data warehouse implemented and loaded in Assignment 1. This warehouse was built from a local snapshot of the dataset “**Crimes - 2001 to Present**”, published by the Chicago Police Department through the City of Chicago Open Data Portal. The selected snapshot contains **759,161 records** and **22 attributes**, covering the period from **1 January 2023 to 1 January 2026**. Each source record represents one unique reported crime incident.

The selected dataset includes operational details such as incident ID, case number, incident date, offence code, offence type, location, arrest status, domestic status, and police administrative area information. This operational source was used in Assignment 1 to design and populate a dimensional data warehouse for analytical reporting.

This dataset was selected to support the business scenario **Urban Public Safety Incident Response and Community Risk Intelligence**. It provides a suitable basis for crime trend analysis, offence pattern analysis, location-based risk analysis, and police-area-based reporting. The dataset is also appropriate for building a Data Warehousing and Business Intelligence solution because it supports dimensional modelling, ETL processing, and analytical querying.

1.1 Original Dataset Overview

Dataset Name: Crimes - 2001 to Present

Provided By: Chicago Police Department

Source: [City of Chicago Open Data Portal](#)

The original dataset contains reported crime incidents that occurred within the City of Chicago. According to the source description used in Assignment 1, the full public dataset contains approximately **8.52 million records** and **22 attributes**, is extracted from the **Citizen Law Enforcement Analysis and Reporting (CLEAR)** system, and is updated daily. Incident addresses are presented only at block level to preserve privacy, while still maintaining analytical usefulness.

For this project, a reduced local snapshot of the dataset was used instead of the

full public version. This snapshot served as the operational source for the warehouse implementation completed in Assignment 1.

1.2 Operational Source Structure

Before developing the warehouse, the source data was analysed at the operational level. The source grain was defined as **one record per reported crime incident**. In the prepared source model, the central operational entity is **Crime Incident**, supported by recurring descriptive entities such as **Case File**, **Offense Definition**, **Coordinate Point**, **District**, **Beat**, **Ward**, and **Community Area**. These were identified because the same business values recur across many incident records.

For example, a case number may be associated with multiple incidents, the IUCR code identifies a reusable offence definition, and geographic and policing codes such as district, beat, ward, and community area are repeated across many incident records. This operational understanding was used as the basis for converting the source into a dimensional warehouse design.

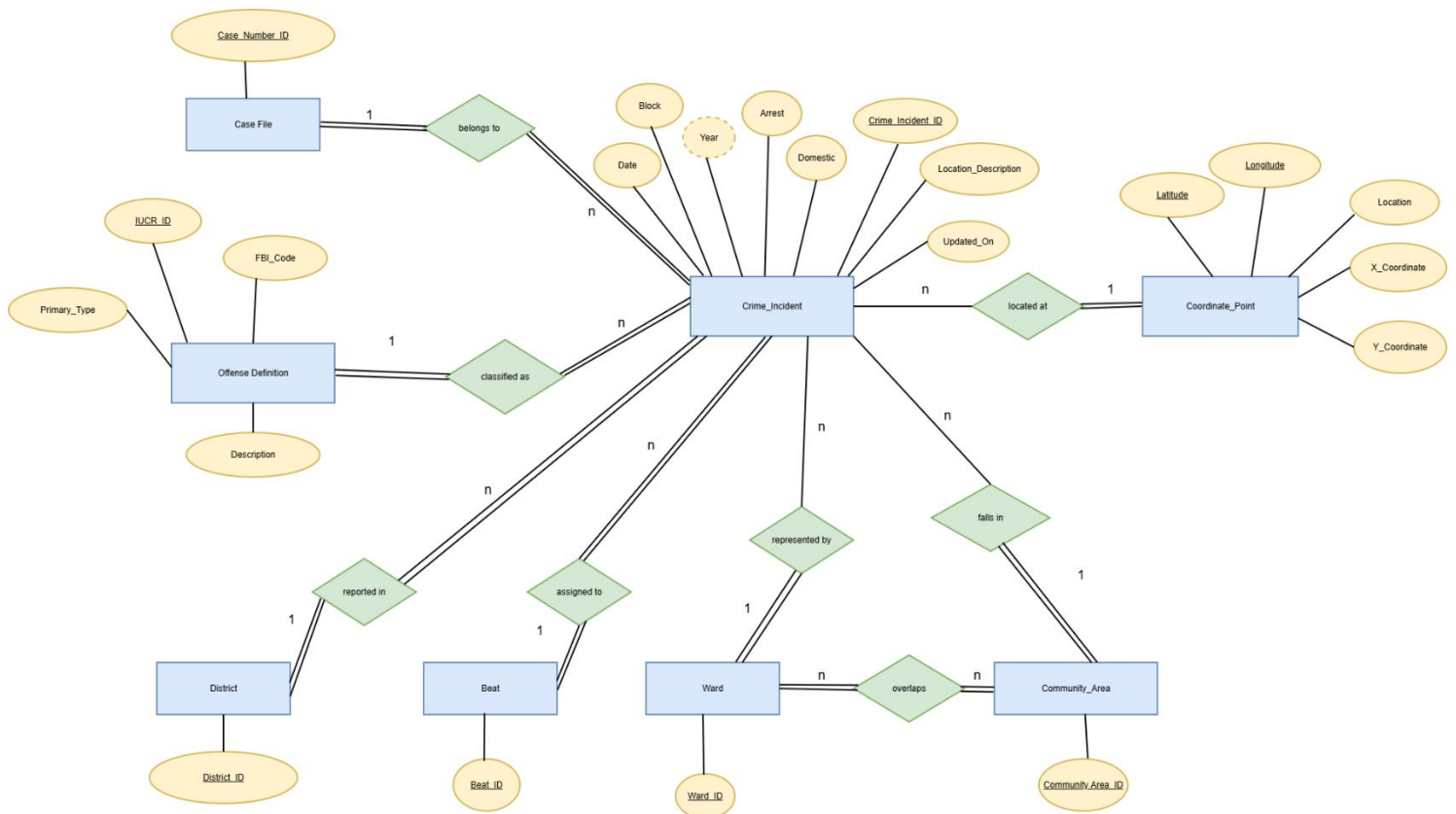


Figure 1: Figure 1: ER Diagram for the Operational Crime Incident Source Structure

1.3 Data Warehouse Used for Assignment 2

The actual data source for Assignment 2 is not the raw crime file itself, but the **data warehouse implemented in Assignment 1**. According to the Assignment 1 design, the warehouse was implemented in SQL Server in the **DWBI_Warehouse** database using a **star schema**. This dimensional design was selected because it provides a simple and efficient structure for analysing measures across multiple descriptive dimensions.

The warehouse consists of:

- One fact table: **FactCrimeIncident**
- Four dimension tables: **DimDate**
- **DimOffense**
- **DimLocation**
- **DimPoliceArea**

This warehouse is the analytical source used in Assignment 2 for reporting, dashboard development, and business intelligence analysis. It enables crime incidents to be examined by time, offence type, location, and police administrative area.

1.4 Fact Table and Grain

The central fact table in the warehouse is **FactCrimeIncident**. The grain of this fact table is defined as:

One row in FactCrimeIncident represents one unique crime incident identified by IncidentID.

The fact table stores foreign keys linking each incident to the dimensions, together with identifiers and numeric analytical measures. Based on the Assignment 1 implementation, the structure includes the following:

Foreign Keys

- DateKey
- OffenseKey
- LocationKey
- PoliceAreaKey

Degenerate / Natural Identifiers

- IncidentID

- CaseNumber

Measures

- IncidentCount
- ArrestCount
- DomesticCount

Audit Attribute

- SourceUpdatedOn

This design allows each incident to be aggregated and analysed from multiple business perspectives while preserving the incident-level detail needed for BI reporting.

1.5 Dimension Tables

The warehouse contains four dimensions that provide descriptive context for the fact table.

1.5.1 DimDate

DimDate supports time-based analysis. It allows incidents to be analysed by date, day, month, quarter, year, and weekend indicator. Important attributes include **FullDate**, **DayName**, **MonthName**, **QuarterNumber**, **YearNumber**, and **WeekendFlag**.

1.5.2 DimOffense

DimOffense supports offence-based analysis. It allows incident reporting by offence code, primary crime category, detailed offence description, and FBI classification. Important attributes include **IUCR**, **PrimaryType**, **OffenseDescription**, and **FBICode**.

1.5.3 DimLocation

DimLocation supports spatial and location-based analysis. It includes both descriptive location values and coordinate information, enabling block-level and coordinate-based crime analysis. Key attributes include **Block**, **LocationDescription**, **Latitude**, **Longitude**, **XCoordinate**, **YCoordinate**, and **LocationText**.

1.5.4 DimPoliceArea

DimPoliceArea supports analysis by police and administrative geography. It enables incidents to be grouped by district, beat, ward, and community area for operational and strategic reporting. This dimension is especially useful for identifying area-level crime concentration and community risk patterns.

1.6 Star Schema of the Data Warehouse

The warehouse follows a **star schema** in which the central fact table, FactCrimeIncident, is directly related to the four dimensions: DimDate, DimOffense, DimLocation, and DimPoliceArea. This structure was selected in Assignment 1 because it simplifies analytical querying and improves reporting efficiency.

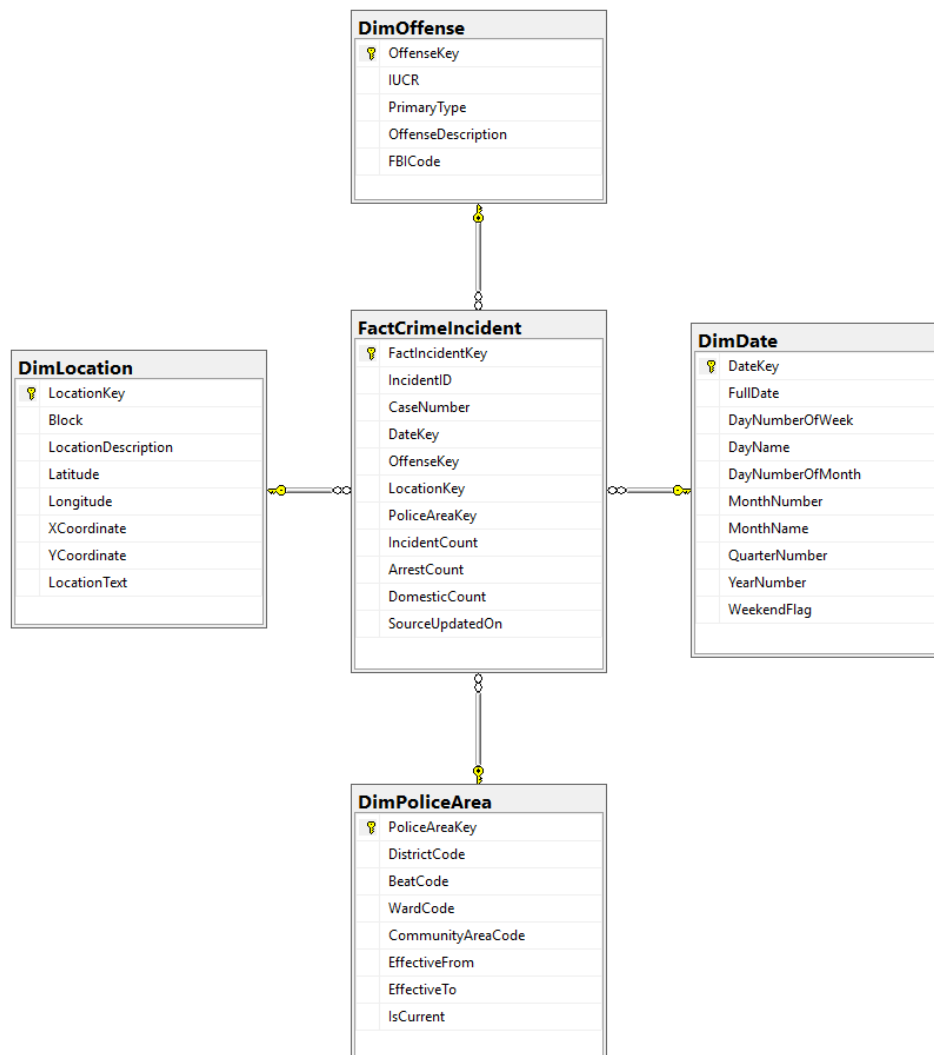


Figure 2: Figure 2: Star Schema of the Crime Data Warehouse

The star schema places FactCrimeIncident at the centre and links it directly to the descriptive dimensions used for time, offence, location, and police-area analysis.

1.7 High Level Design Architecture

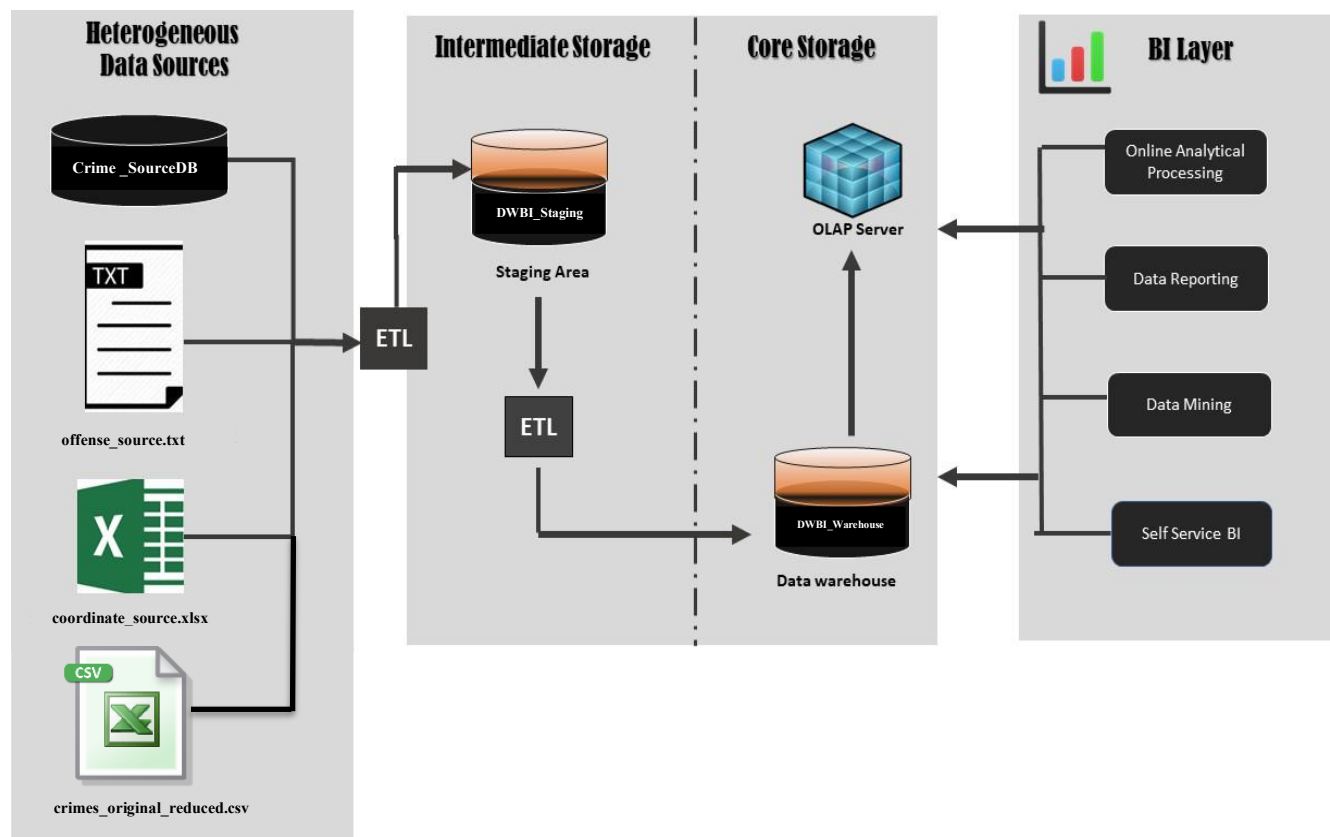


Figure 3: Layered Architecture of the Crime Data Warehouse System

The figure above illustrates the high-level architecture of the proposed crime data warehouse and business intelligence solution. Data is collected from multiple heterogeneous sources, including the Crime_SourceDB source database, offense_source.txt, coordinate_source.xlsx, and crimes_original_reduced.csv. Since real-time data capture is not required, the raw source data is first extracted into the DWBI_Staging database through SSIS.

At the staging layer, ETL operations such as data cleansing, data conversion, lookup processing, and derived-column transformations are carried out before loading the data into the warehouse. The processed data is then stored in DWBI_Warehouse, which contains the dimensional model used for analysis.

Finally, the BI layer supports reporting, OLAP-based analysis, data mining, and self-service B

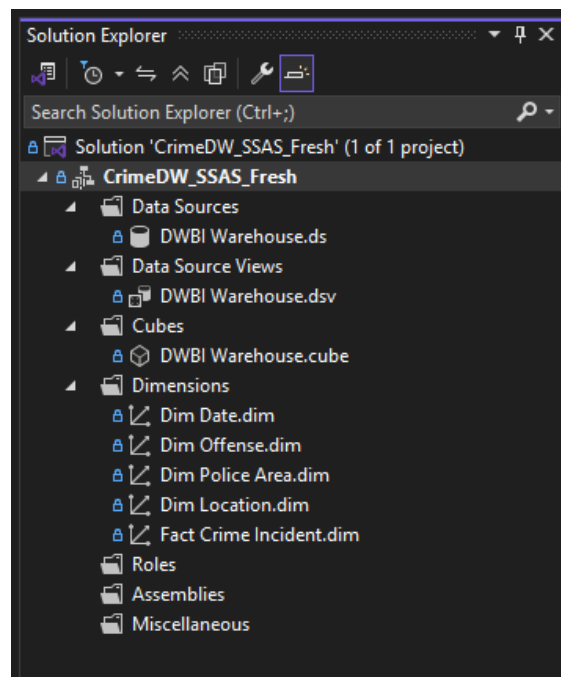
Step 2: SSAS Cube implementation

2.1 Introduction

In this step, an **SSAS Multidimensional cube** was created by using the **DWBI_Warehouse** database as the main data source. The purpose of the cube was to support crime analysis by organizing fact data and related dimensions in a multidimensional model. The cube was designed with the required measures from the fact table and connected to the relevant dimensions. Hierarchies were also implemented to support drill-down analysis.

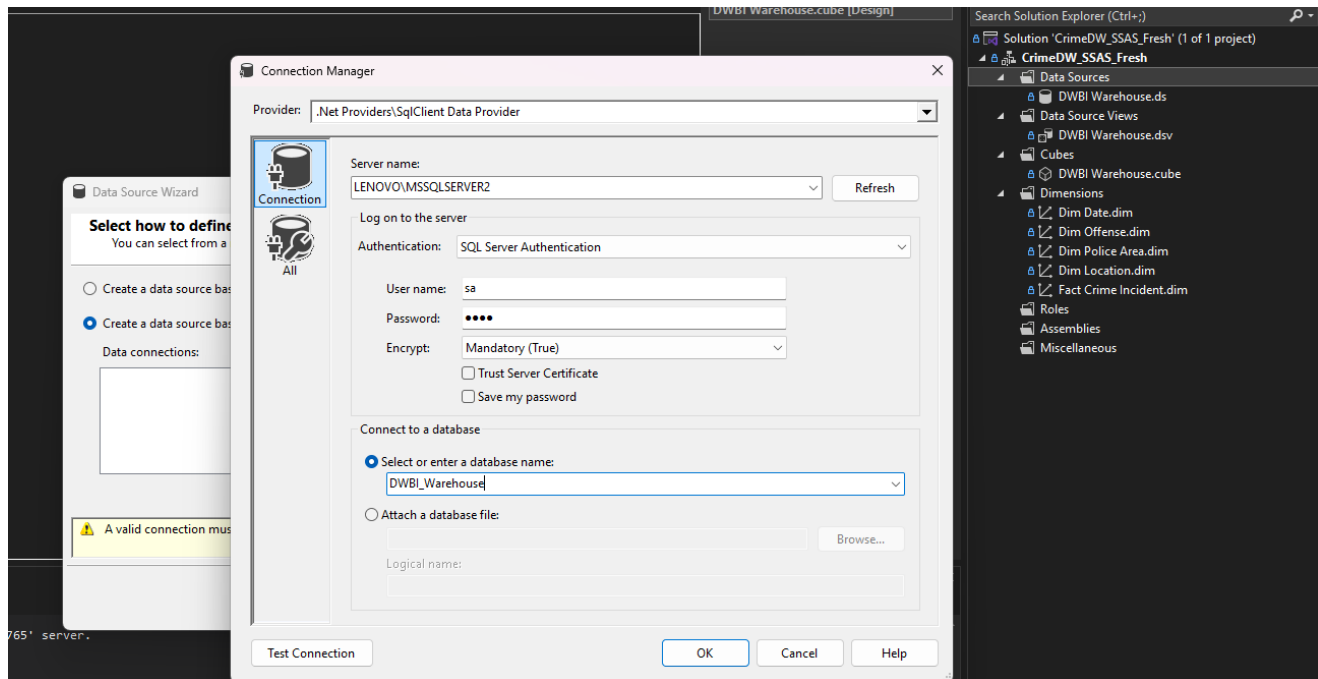
2.2 Creation of the SSAS Project

A new **Analysis Services Multidimensional Project** was created in Visual Studio and named **CrimeDW_SSAS_Fresh**. This project was used to design, deploy, process, and browse the cube.



2.3 Data Source Configuration

A **Data Source** was created and connected to the **DWBI_Warehouse** database. This data warehouse was used as the source for the SSAS cube. The data source was configured so that SSAS could access the warehouse tables during deployment and processing.

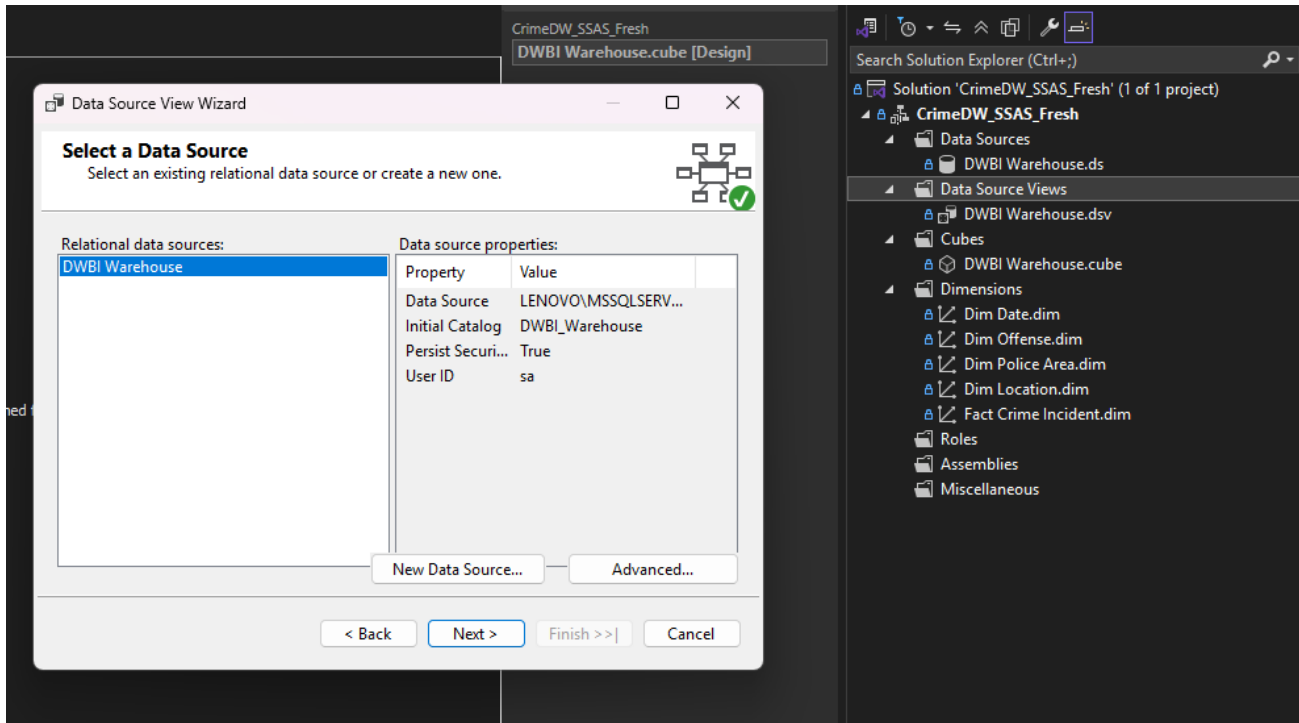


2.4 Data Source View (DSV)

A **Data Source View** was created to define the tables required for the cube. The following tables were included:

- **FactCrimeIncident**
- **DimDate**
- **DimOffense**
- **DimLocation**
- **DimPoliceArea**

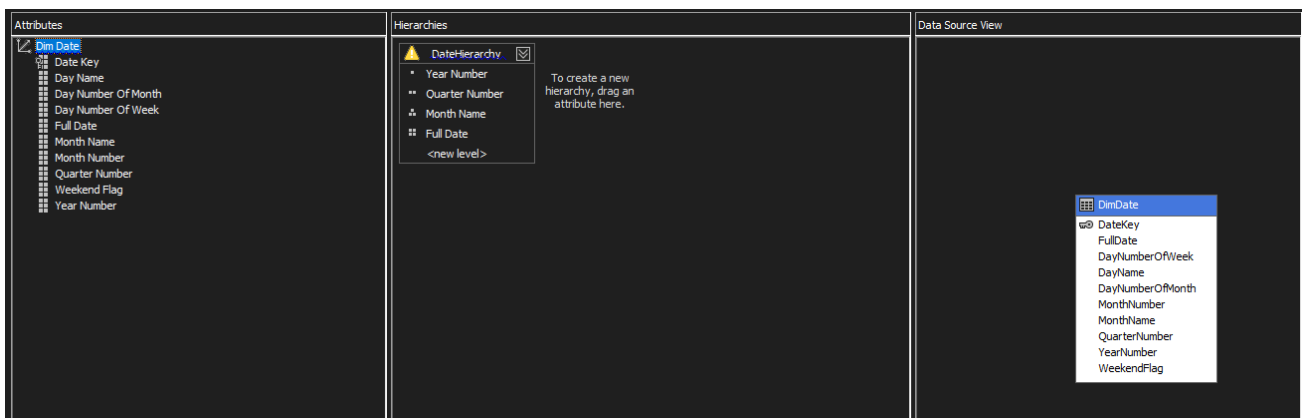
The Data Source View was used as the logical layer between the warehouse and the cube. It also provided the table relationships required to link the fact table with the dimensions.

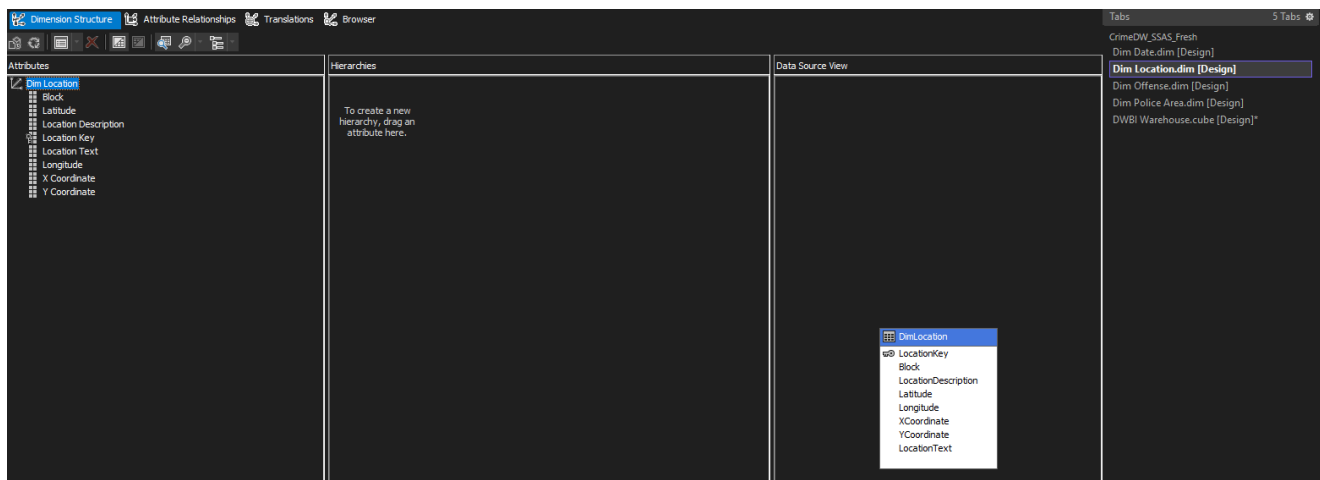
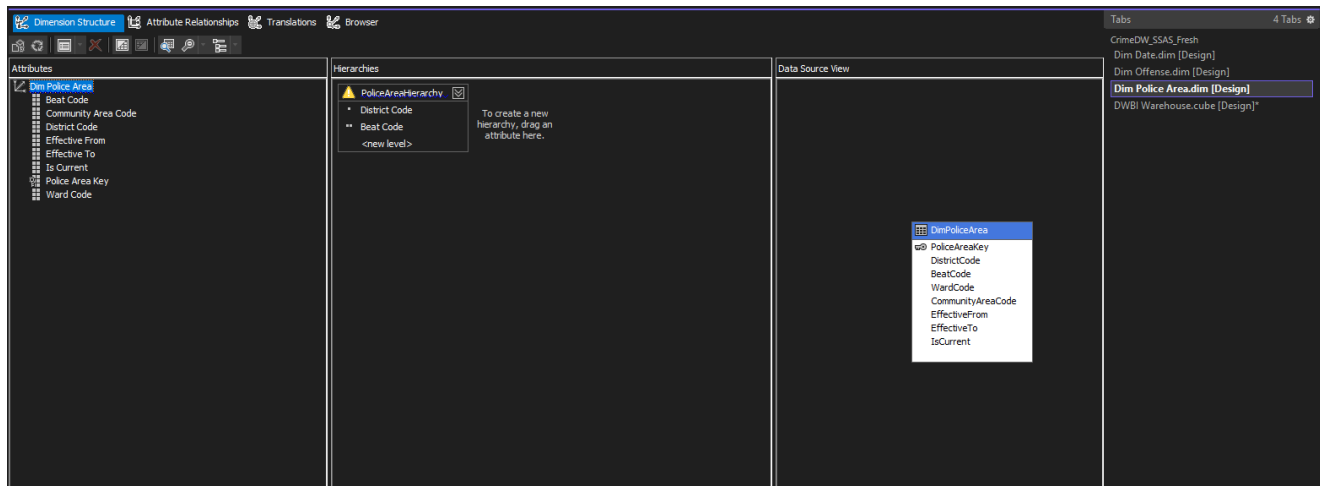
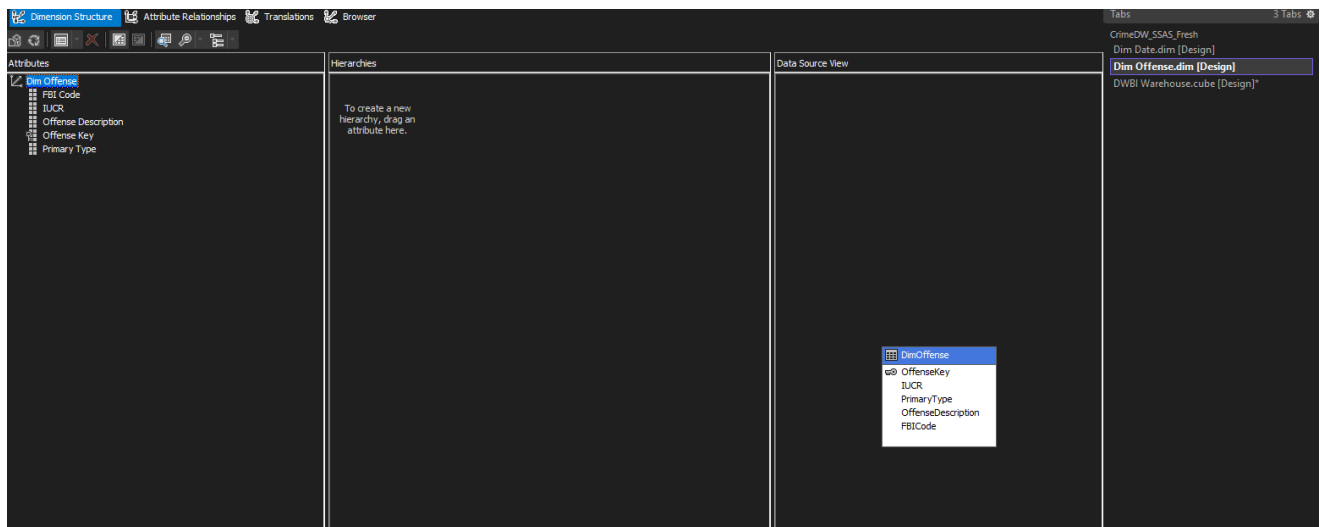


2.5 Dimension Creation

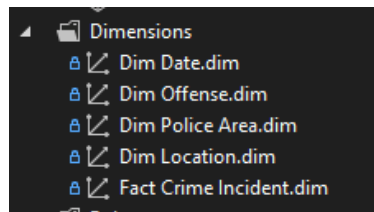
The following dimensions were created from the Data Source View:

- **DimDate**
- **DimOffense**
- **DimLocation**
- **DimPoliceArea**





These dimensions were used to provide different perspectives for analyzing the fact data.



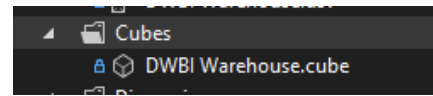
2.6 Cube Creation

A new cube was created by using **FactCrimeIncident** as the **fact table / measure group**. The cube was designed to analyze crime incidents by date, offense, location, and police area.

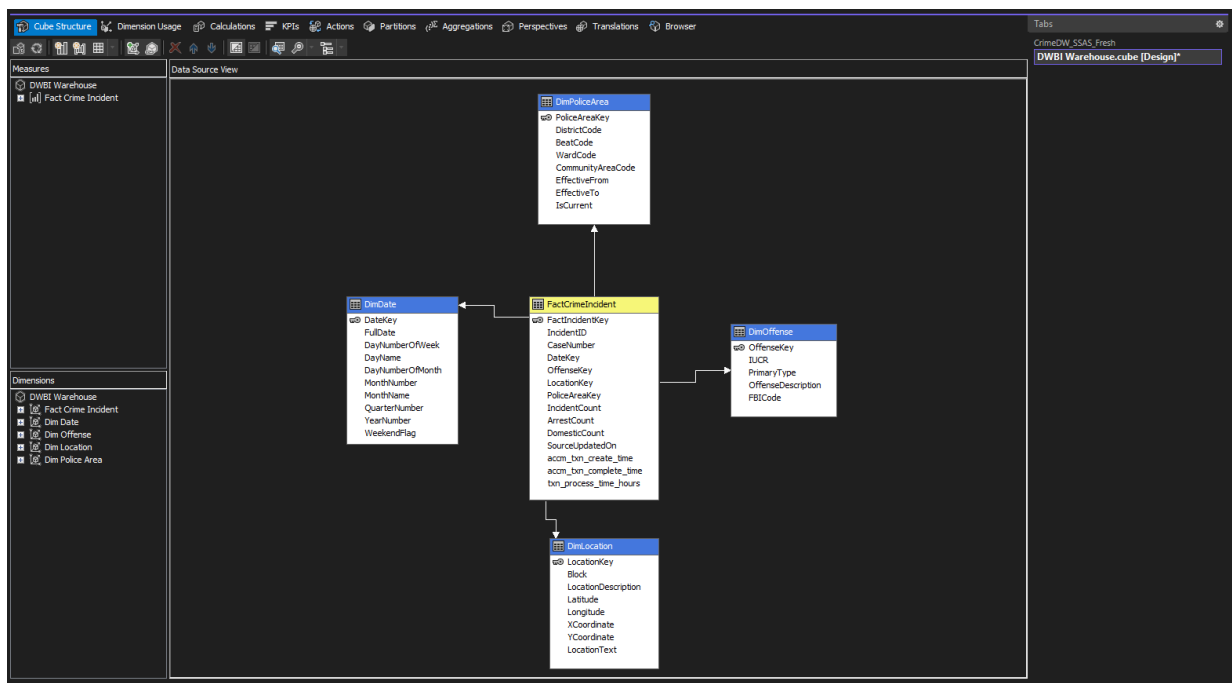
Measures included in the cube

The following measures were selected from the fact table:

- **IncidentCount**
- **ArrestCount**
- **DomesticCount**



These measures were included because they are the main numerical values required for analysis in the assignment.

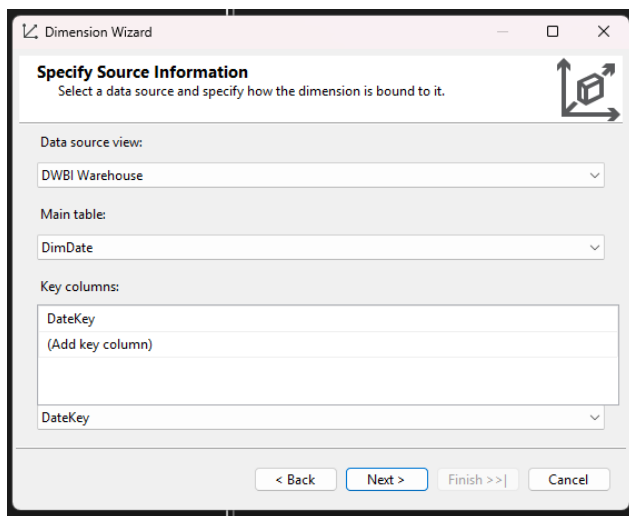


2.7 Dimension Usage

The cube was connected to the dimensions through the **Dimension Usage** tab. The relationships were configured as follows:

- **FactCrimeIncident.DateKey** → **DimDate.DateKey**
- **FactCrimeIncident.OffenseKey** → **DimOffense.OffenseKey**
- **FactCrimeIncident.LocationKey** → **DimLocation.LocationKey**
- **FactCrimeIncident.PoliceAreaKey** → **DimPoliceArea.PoliceAreaKey**

This ensured that the cube measures could be analyzed correctly by the connected dimensions.



Dimension Wizard

Specify Source Information
Select a data source and specify how the dimension is bound to it.

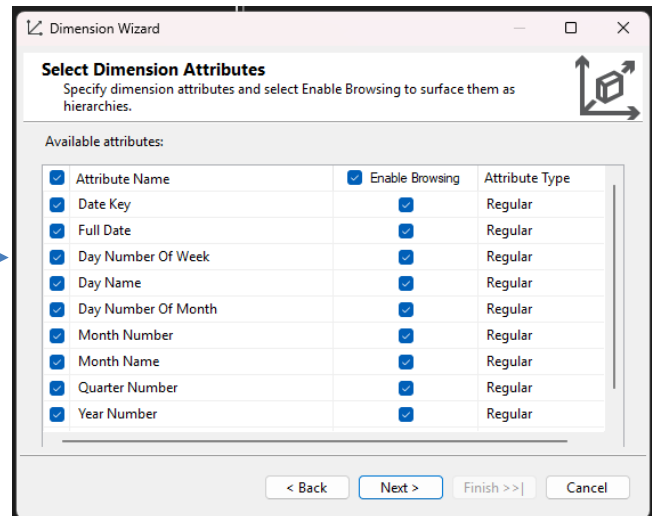
Data source view:
DWBI Warehouse

Main table:
DimDate

Key columns:
DateKey
(Add key column)

DateKey

< Back Next > Finish >>| Cancel



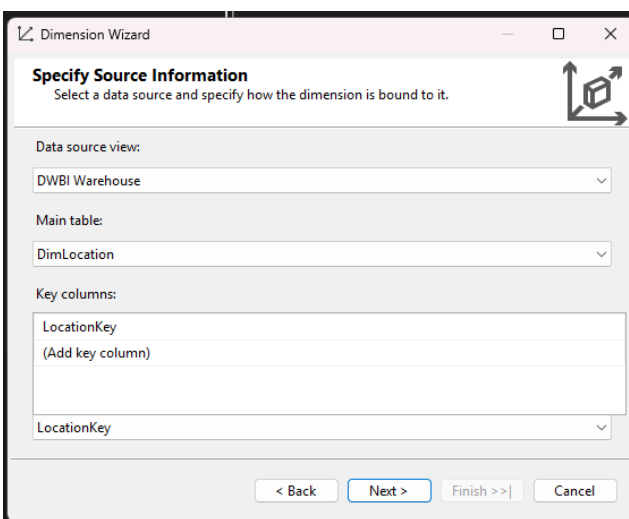
Dimension Wizard

Select Dimension Attributes
Specify dimension attributes and select Enable Browsing to surface them as hierarchies.

Available attributes:

Attribute Name	Enable Browsing	Attribute Type
Date Key	☒	Regular
Full Date	☒	Regular
Day Number Of Week	☒	Regular
Day Name	☒	Regular
Day Number Of Month	☒	Regular
Month Number	☒	Regular
Month Name	☒	Regular
Quarter Number	☒	Regular
Year Number	☒	Regular

< Back Next > Finish >>| Cancel



Dimension Wizard

Specify Source Information
Select a data source and specify how the dimension is bound to it.

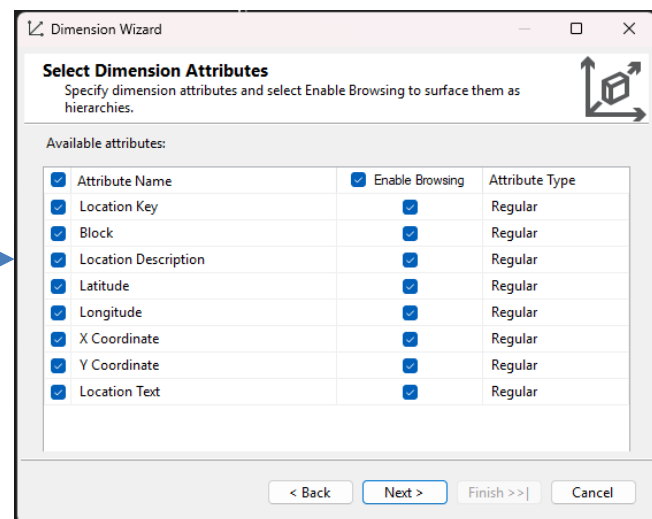
Data source view:
DWBI Warehouse

Main table:
DimLocation

Key columns:
LocationKey
(Add key column)

LocationKey

< Back Next > Finish >>| Cancel



Dimension Wizard

Select Dimension Attributes
Specify dimension attributes and select Enable Browsing to surface them as hierarchies.

Available attributes:

Attribute Name	Enable Browsing	Attribute Type
Location Key	☒	Regular
Block	☒	Regular
Location Description	☒	Regular
Latitude	☒	Regular
Longitude	☒	Regular
X Coordinate	☒	Regular
Y Coordinate	☒	Regular
Location Text	☒	Regular

< Back Next > Finish >>| Cancel

Dimension Wizard

Specify Source Information
Select a data source and specify how the dimension is bound to it.

Data source view:
DWBI Warehouse

Main table:
DimOffense

Key columns:
OffenseKey
(Add key column)
OffenseKey

< Back Next > Finish >>| Cancel

Dimension Wizard

Select Dimension Attributes
Specify dimension attributes and select Enable Browsing to surface them as hierarchies.

Available attributes:

Attribute Name	Enable Browsing	Attribute Type
Offense Key	☒	Regular
IUCR	☒	Regular
Primary Type	☒	Regular
Offense Description	☒	Regular
FBI Code	☒	Regular

< Back Next > Finish >>| Cancel

Dimension Wizard

Specify Source Information
Select a data source and specify how the dimension is bound to it.

Data source view:
DWBI Warehouse

Main table:
DimPoliceArea

Key columns:
PoliceAreaKey
(Add key column)
PoliceAreaKey

< Back Next > Finish >>| Cancel

Dimension Wizard

Select Dimension Attributes
Specify dimension attributes and select Enable Browsing to surface them as hierarchies.

Available attributes:

Attribute Name	Enable Browsing	Attribute Type
Police Area Key	☒	Regular
District Code	☒	Regular
Beat Code	☒	Regular
Ward Code	☒	Regular
Community Area Code	☒	Regular
Effective From	☒	Regular
Effective To	☒	Regular
Is Current	☒	Regular

< Back Next > Finish >>| Cancel

2.8 Hierarchies Implemented

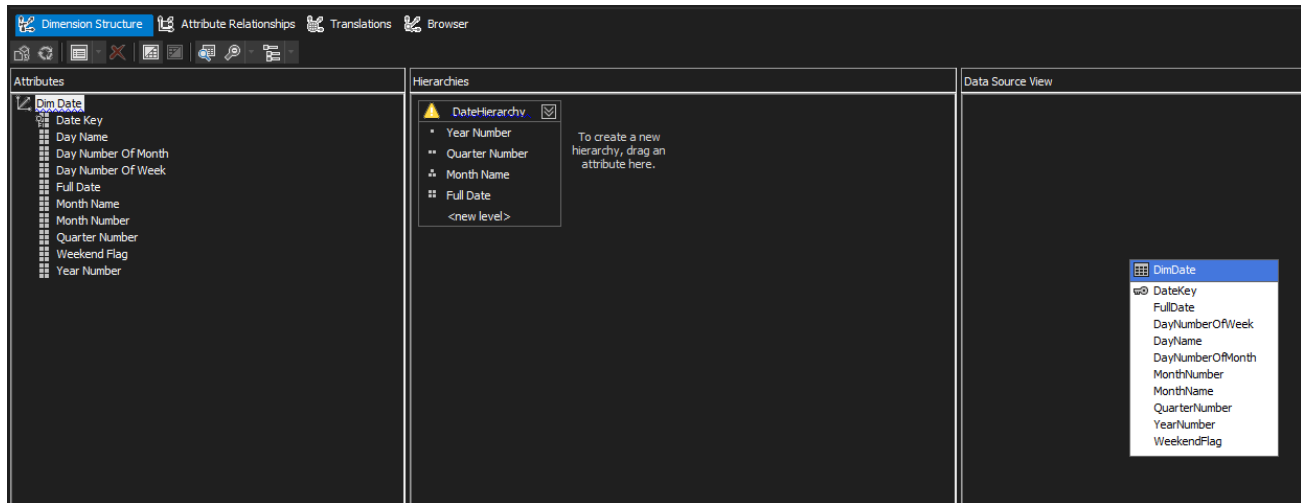
In this implementation, **two hierarchies** were created.

1. DateHierarchy

The first hierarchy was created in the **DimDate** dimension:

- **Year Number**
- **Quarter Number**
- **Month Name**
- **Full Date**

This hierarchy allows drill-down analysis from year level down to full date level.

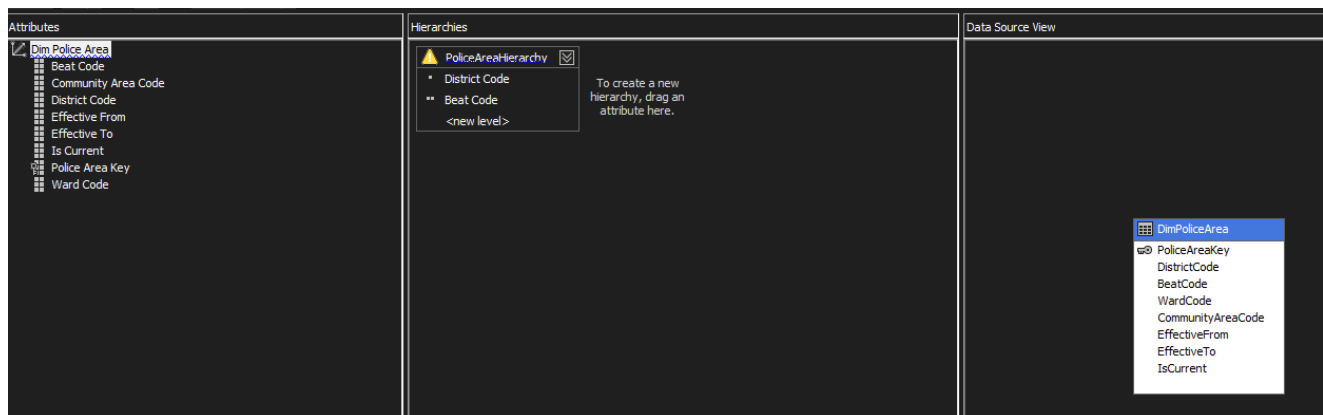


2. PoliceAreaHierarchy

The second hierarchy was created in the **DimPoliceArea** dimension:

- **District Code**
- **Beat Code**

This hierarchy allows drill-down analysis from district level to beat level.

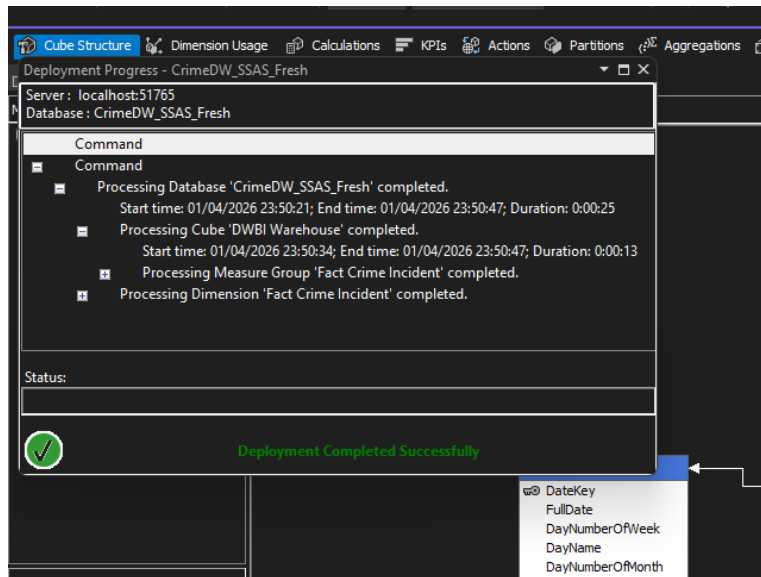


2.9 Deployment and Processing

After the cube design was completed, the project was successfully **deployed** to SQL Server Analysis Services. The database, cube, measure group, and dimensions were then **processed** successfully.

Deployment and processing confirmed that:

- The data source was connected correctly,
- The dimensions were linked correctly,
- The cube structure was valid,
- And the hierarchy implementation worked as expected.



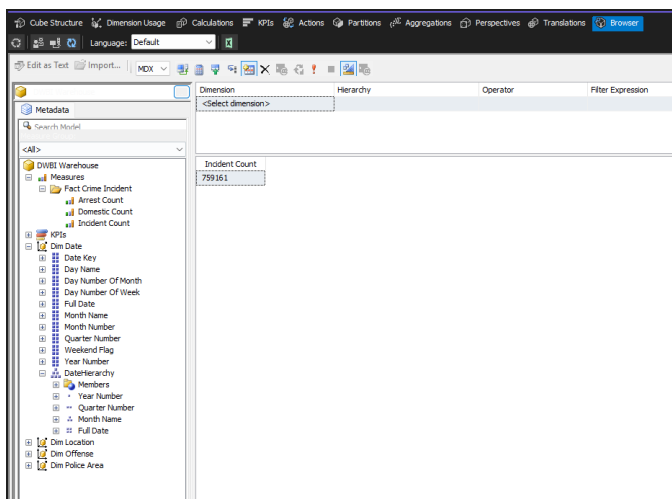
2.10 Cube Validation Using Browser

The cube was validated using the **Browser** tab in SSAS. The measures were tested across the connected dimensions and hierarchies.

Examples of analysis performed in the browser include:

- **IncidentCount** by **DateHierarchy**
- **ArrestCount** by offense-related attributes
- **DomesticCount** by **PoliceAreaHierarchy**

The browser results confirmed that the cube was functioning correctly and could be used for multidimensional analysis.



Year Number	Quarter Number	Month Name	Full Date	Incident Count
2023	1	February	2/1/2023	659
2023	1	February	2/2/2023	648
2023	1	February	2/3/2023	595
2023	1	February	2/4/2023	684
2023	1	February	2/5/2023	664
2023	1	February	2/6/2023	805
2023	1	February	2/7/2023	656
2023	1	February	2/8/2023	818
2023	1	February	2/9/2023	819
2023	1	February	2/10/2023	790
2023	1	February	2/11/2023	770
2023	1	February	2/12/2023	720
2023	1	February	2/13/2023	660
2023	1	February	2/14/2023	688
2023	1	February	2/15/2023	627
2023	1	February	2/16/2023	608
2023	1	February	2/17/2023	681
2023	1	February	2/18/2023	670
2023	1	February	2/19/2023	731
2023	1	February	2/20/2023	636
2023	1	February	2/21/2023	603
2023	1	February	2/22/2023	555
2023	1	February	2/23/2023	652
2023	1	February	2/24/2023	628
2023	1	February	2/25/2023	646
2023	1	February	2/26/2023	672
2023	1	February	2/27/2023	637

Dimension	Hierarchy	Operator	Filter Expression
<Select dimension>			

Year Number	Quarter Number	Month Name	Full Date	Incident Count	Domestic Count
2023	1	February	2/1/2023	659	144
2023	1	February	2/2/2023	648	95
2023	1	February	2/3/2023	595	111
2023	1	February	2/4/2023	684	140
2023	1	February	2/5/2023	664	112
2023	1	February	2/6/2023	805	152
2023	1	February	2/7/2023	656	155
2023	1	February	2/8/2023	818	113
2023	1	February	2/9/2023	819	126
2023	1	February	2/10/2023	790	147
2023	1	February	2/11/2023	770	117
2023	1	February	2/12/2023	720	137
2023	1	February	2/13/2023	660	119
2023	1	February	2/14/2023	688	135
2023	1	February	2/15/2023	627	129
2023	1	February	2/16/2023	608	116
2023	1	February	2/17/2023	681	114
2023	1	February	2/18/2023	670	119
2023	1	February	2/19/2023	731	152
2023	1	February	2/20/2023	636	110
2023	1	February	2/21/2023	603	101
2023	1	February	2/22/2023	555	98
2023	1	February	2/23/2023	652	94
2023	1	February	2/24/2023	628	106
2023	1	February	2/25/2023	646	125
2023	1	February	2/26/2023	672	135
2023	1	February	2/27/2023	637	138
2023	1	February	2/28/2023	677	104
2023	1	January	1/1/2023	1130	283
2023	1	January	1/2/2023	726	123
2023	1	January	1/3/2023	778	139

Dimension	Hierarchy	Operator	Filter Expression
<Select dimension>			

Year Number	Quarter Number	Month Name	Full Date	District Code	Beat Code	Incident Count	Domestic Count
2023	1	February	2/1/2023	111	3	0	
2023	1	February	2/1/2023	112	4	0	
2023	1	February	2/1/2023	113	1	0	
2023	1	February	2/1/2023	121	1	1	
2023	1	February	2/1/2023	122	6	1	
2023	1	February	2/1/2023	123	10	2	
2023	1	February	2/1/2023	131	6	0	
2023	1	February	2/1/2023	132	1	0	
2023	1	February	2/1/2023	133	3	0	
2023	1	February	2/1/2023	1011	4	0	
2023	1	February	2/1/2023	1012	2	1	
2023	1	February	2/1/2023	1013	3	1	
2023	1	February	2/1/2023	1014	2	1	
2023	1	February	2/1/2023	1021	5	2	
2023	1	February	2/1/2023	1024	3	0	
2023	1	February	2/1/2023	1032	2	0	
2023	1	February	2/1/2023	1033	1	0	
2023	1	February	2/1/2023	1034	2	1	
2023	1	February	2/1/2023	1111	2	0	
2023	1	February	2/1/2023	1112	3	0	
2023	1	February	2/1/2023	1113	3	0	
2023	1	February	2/1/2023	1115	3	0	
2023	1	February	2/1/2023	1121	1	0	
2023	1	February	2/1/2023	1122	2	0	
2023	1	February	2/1/2023	1123	3	1	
2023	1	February	2/1/2023	1124	1	0	
2023	1	February	2/1/2023	1125	3	0	
2023	1	February	2/1/2023	1131	3	2	
2023	1	February	2/1/2023	1132	6	2	

Step 3: Demonstration of OLAP operations

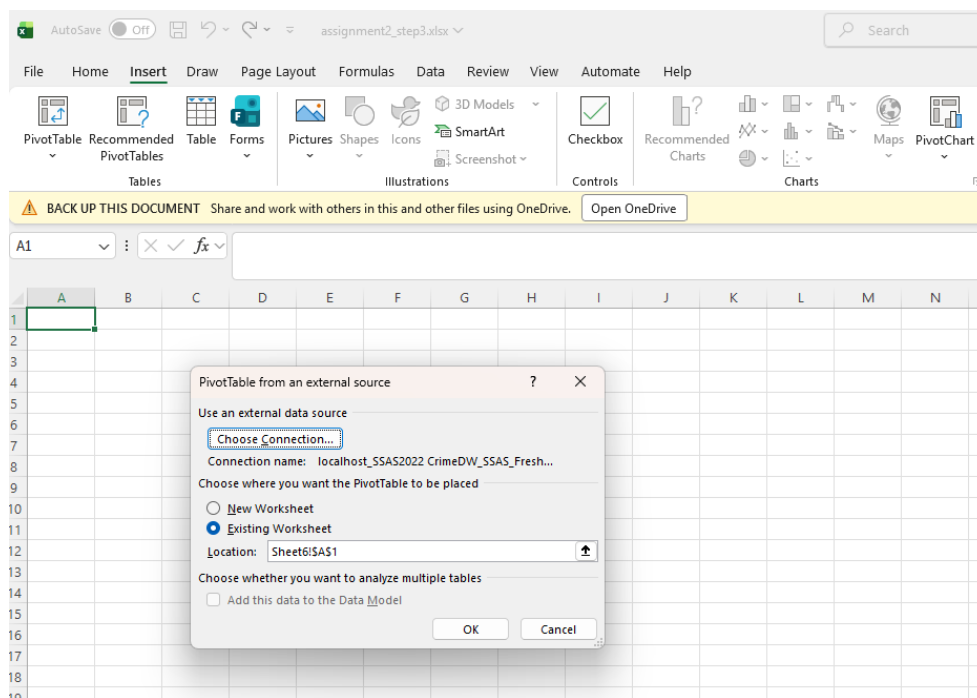
3.1 Introduction

In this step, an Excel workbook was connected to the SSAS cube in order to demonstrate the main **OLAP operations**. PivotTables and PivotCharts were created from the deployed cube so that the multidimensional analysis features could be shown clearly. The OLAP operations demonstrated were **Roll-up, Drill-down, Slice, Dice, and Pivot**.

3.2 Connecting Excel to the Cube

A new Excel workbook was created and connected to the deployed SSAS cube. The connection was created by using the **Data** tab in Excel and connecting to the Analysis Services cube. The server used for the connection was the deployed SSAS instance, and the cube selected was **Cube_CrimeIncident** from the **CrimeDW_SSAS_Fresh** database.

After the connection was established, the cube fields became available in the **PivotTable Fields** pane. These fields included the measures from the fact table and the hierarchies/attributes from the dimensions such as **DimDate**, **DimOffense**, **DimLocation**, and **DimPoliceArea**.



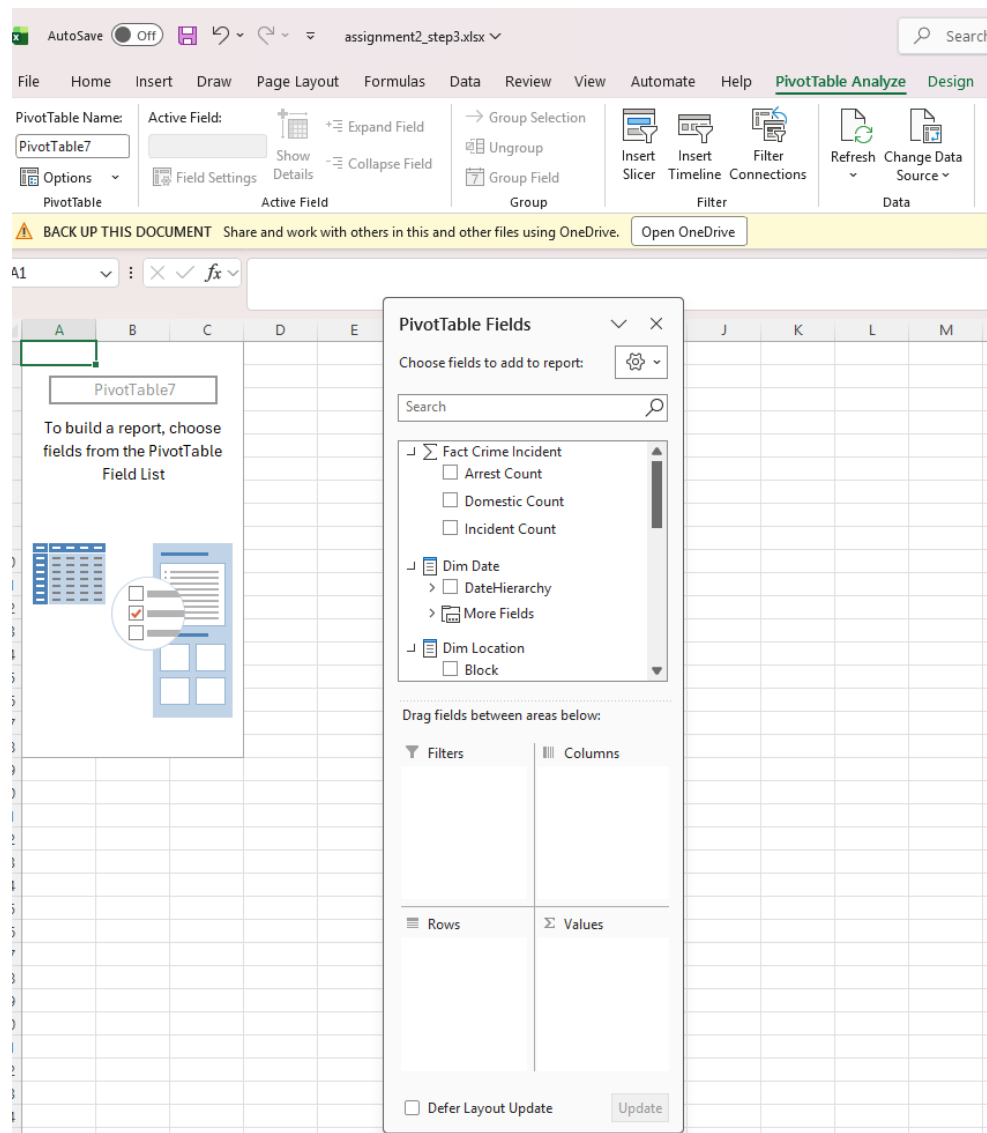
3.3 Excel Visualizations Created

Separate PivotTable sheets were created in Excel to demonstrate each OLAP operation. The measures mainly used in the analysis were:

- **Incident Count**
- **Arrest Count**
- **Domestic Count**

The dimension fields and hierarchies used in the analysis included:

- **DateHierarchy** from **DimDate**
- **Primary Type** from **DimOffense**
- **PoliceAreaHierarchy** from **DimPoliceArea**



3.4 Roll-up Operation

The **Roll-up** operation was demonstrated by summarizing crime incidents at a higher level of the date hierarchy.

Configuration used

- **Values:** Incident Count
- **Rows:** DateHierarchy

The PivotTable showed the total number of crime incidents summarized by **Year**. Since the detailed data was aggregated to a higher level, this represented the **Roll-up** operation.

The screenshot shows an Excel PivotTable with the following data:

Row Labels	Incident Count
2023	263291
2024	
1	63008
2	66681
3	67637
4	61708
2025	236788
2026	48
Grand Total	759161

The PivotTable Fields task pane is open on the right, showing the following configuration:

- Choose fields to add to report:** Fact Crime Incident (Arrest Count, Domestic Count, Incident Count), Dim Date (DateHierarchy, More Fields), Dim Location (Block).
- Drag fields between areas below:**
 - Filters:**
 - Columns:**
 - Rows:** DateHierarchy
 - Values:** Incident Count
- Defer Layout Update:** (unchecked)
- Update:** (button)

Explanation

Roll-up is used to move from detailed data to summarized data. In this case, crime incidents were aggregated at the year level, which allowed high-level trend analysis.

3.5 Drill-down Operation

The **Drill-down** operation was demonstrated by expanding the date hierarchy in the PivotTable.

Configuration used

- **Values:** Incident Count
- **Rows:** DateHierarchy

The year level was expanded to show lower hierarchy levels such as **Quarter**, **Month**, and **Date**.

Row Labels	Incident Count
2023	
1	
February	18999
January	23142
March	21800
2	
April	21237
June	22313
May	22602
3	
August	22831
July	23275
September	22047
4	
December	21235
November	20979
October	22831
2024	259034
2025	236788
2026	48
Grand Total	759161

PivotTable Fields

Choose fields to add to report:

Search

☒ Fact Crime Incident

☐ Arrest Count

☐ Domestic Count

☒ Incident Count

☒ Dim Date

☒ DateHierarchy

☐ More Fields

☒ Dim Location

☐ Block

Drag fields between areas below:

Filters

Columns

Rows

DateHierarchy

Values

Incident Count

☐ Defer Layout Update

Update

Explanation

Drill-down is the opposite of roll-up. It allows the user to move from summarized data to more detailed data. In this cube, the DateHierarchy allowed navigation from the year level to lower levels for detailed analysis.

3.6 Slice Operation

The **Slice** operation was demonstrated by filtering the cube using a **single dimension value**.

Configuration used

- **Values:** Arrest Count
- **Rows:** DateHierarchy
- **Filter:** Primary Type

A single offense category was selected in the filter, and the PivotTable displayed only the arrest counts related to that selected value.

The screenshot displays a PivotTable and its configuration pane. The PivotTable shows arrest counts for various years (2023, 2024, 2025, 2026) and a Grand Total. The configuration pane shows the PivotTable Fields task pane with the following settings:

- Fact Crime Incident:** Arrest Count (checked), Domestic Count (unchecked), Incident Count (unchecked).
- Dim Date:** DateHierarchy (checked), More Fields (unchecked).
- Dim Location:** (unchecked).
- Filters:** Primary Type (selected).
- Columns:** (empty).
- Rows:** DateHierarchy (selected).
- Values:** Arrest Count (selected).

The PivotTable data is as follows:

Primary Type	Arrest Count
2023	18753
2024	20310
2025	22078
2026	2
Grand Total	61143

Explanation

Slice reduces the cube to one subset by selecting one specific value from one dimension. In this case, one offense type was selected, and only the related arrests were analyzed.

3.7 Dice Operation

The **Dice** operation was demonstrated by filtering the cube using **multiple dimensions and multiple conditions**.

Configuration used

- **Values:** Domestic Count
- **Rows:** PoliceAreaHierarchy
- **Filters:** Year Number, Offense Description, and District Code

Multiple filter values were applied at the same time in order to analyze a smaller sub-cube of data.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Offense Description	\$500 AND UNDER											
2	Year Number	2023											
3	District Code	(Multiple Items)											
4													
5	Row Labels	Domestic Count											
6	1	29											
7	10	72											
8	11	92											
9	12	55											
10	14	25											
11	15	88											
12	16	26											
13	Grand Total	387											
14													
15													
16													
17													
18													
19													
20													
21													
22													
23													
24													
25													
26													
27													
28													
29													
30													
31													
32													
33													

PivotTable Fields

Choose fields to add to report:

Search

Fact Crime Incident

☐ Arrest Count
 ☒ Domestic Count
 ☐ Incident Count

Dim Date

☐ DateHierarchy
 ☐ More Fields

Drag fields between areas below:

Filters

Offense Descri...

Year Number

District Code

Columns

Rows

PoliceAreaHie...

Values

Domestic Count

☐ Defer Layout Update
 Update

Year Number

2023

1

1900

2024

2025

2026

Offense Descript...

\$500 AND UNDER

ABUSE / NEGLECT - ...

AGG CRIM SEX ABUS...

AGG. DOMESTIC BA...

AGG. PROTECTED E...

AGGRAVATED

AGGRAVATED - HAN...

AGGRAVATED - HAN...

District Code

1

10

11

12

14

15

16

-1

Explanation

Dice allows the user to analyze data by selecting multiple values from multiple dimensions. In this implementation, the cube was filtered simultaneously by year, offense type, and police area, producing a more focused analytical view.

3.8 Pivot Operation

The **Pivot** operation was demonstrated by changing the orientation of the dimensions in the PivotTable.

Configuration used

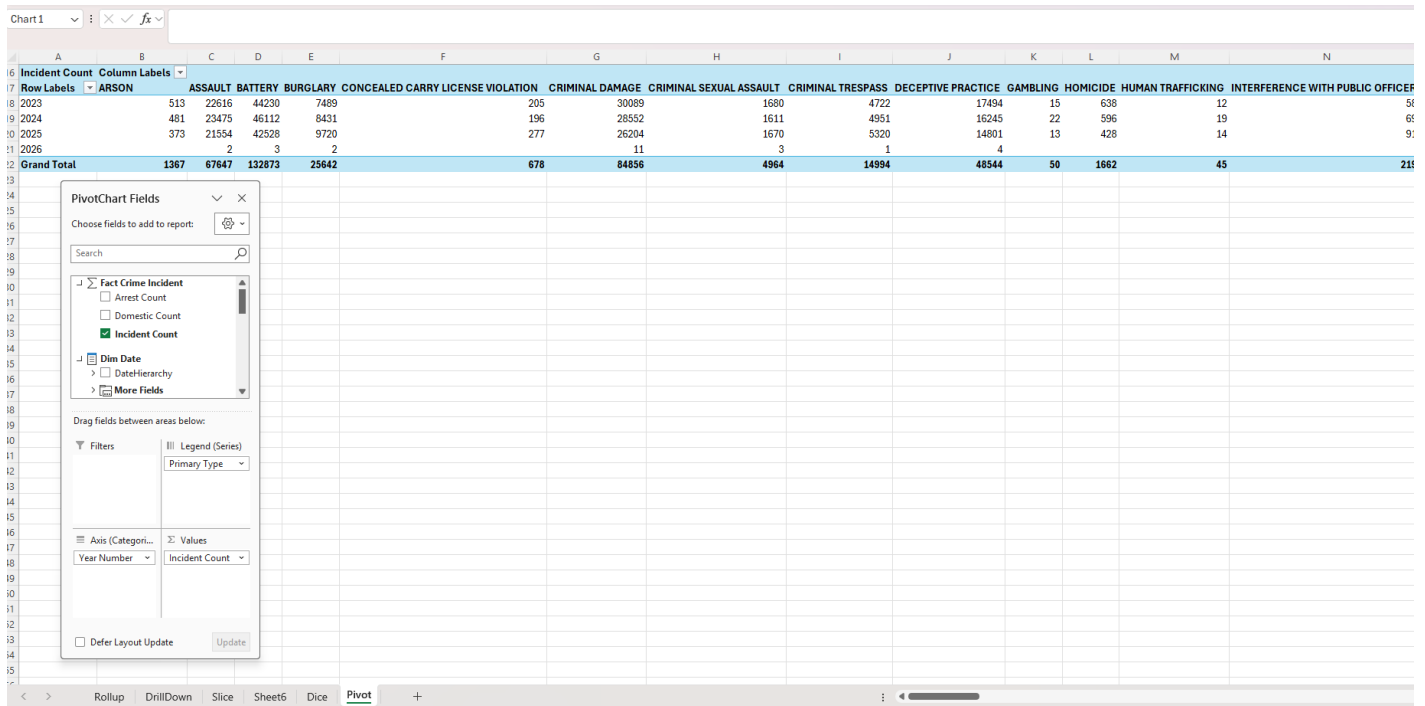
Initial layout:

- **Rows:** Primary Type
- **Columns:** Year Number
- **Values:** Incident Count

Incident Count	Column Labels				
Row Labels	2023	2024	2025	2026	Grand Total
ARSON	513	481	373		1367
ASSAULT	22616	23475	21554	2	67647
BATTERY	44230	46112	42528	3	132873
BURGLARY	7489	8431	9720	2	25642
CONCEALED CARRY LICENSE VIOLATION	205	196	277		678
CRIMINAL DAMAGE	30089	28552	26204	11	84856
CRIMINAL SEXUAL ASSAULT	1680	1611	1670	3	4964
CRIMINAL TRESPASS	4722	4951	5320	1	14994
DECEPTIVE PRACTICE	17494	16245	14801	4	48544
GAMBLING	15	22	13		50
HOMICIDE	638	596	428		1662
HUMAN TRAFFICKING	12	19	14		45
INTERFERENCE WITH PUBLIC OFFICER	586	690	916		2192
INTIMIDATION	237	164	143		544
KIDNAPPING	136	97	88		321
LIQUOR LAW VIOLATION	186	196	200		582
MOTOR VEHICLE THEFT	29255	21709	17234	2	68200
NARCOTICS	5455	5988	7416		18859
NON-CRIMINAL	3	3	3		9
OBSCENITY	42	62	55		159
OFFENSE INVOLVING CHILDREN	1751	1689	1545	3	4988
OTHER NARCOTIC VIOLATION	2	5	10		17
OTHER OFFENSE	15729	17125	16751	4	49609
PROSTITUTION	209	306	189		704
PUBLIC INDECENCY	6	9	10		25
PUBLIC PEACE VIOLATION	862	979	1056		2897
ROBBERY	11040	9105	5812	1	25958
SEX OFFENSE	1361	1247	1302	1	3911
STALKING	531	517	572		1620
THEFT	57489	60493	55037	9	173028
Unknown	102	124	104		330
WEAPONS VIOLATION	8606	7835	5443	2	21886
Grand Total	322364	350034	322788	48	750144

Then the layout was changed to:

- **Rows:** Year Number
- **Columns:** Primary Type
- **Values:** Incident Count



Explanation

Pivot changes the arrangement of data between rows and columns so that the same information can be viewed from a different perspective. This helps users analyze cube data in alternative layouts.

3.9 Use of PivotCharts

To improve visualization, **PivotCharts** were also created from the PivotTables. These charts updated dynamically whenever hierarchy expansion, filters, or row/column arrangements were changed.

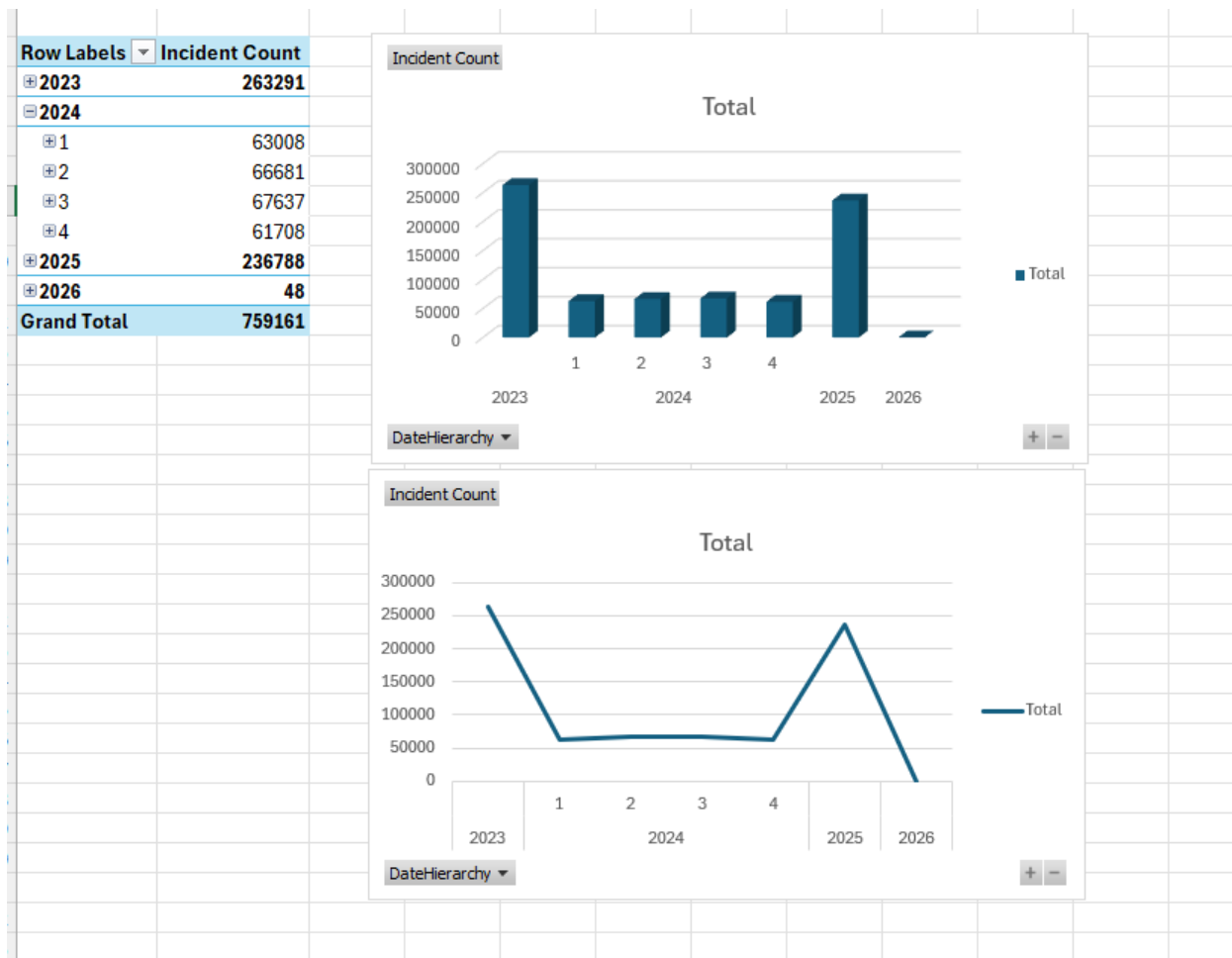
Examples:

- A **Column Chart** was used to visualize roll-up and drill-down on the date hierarchy.

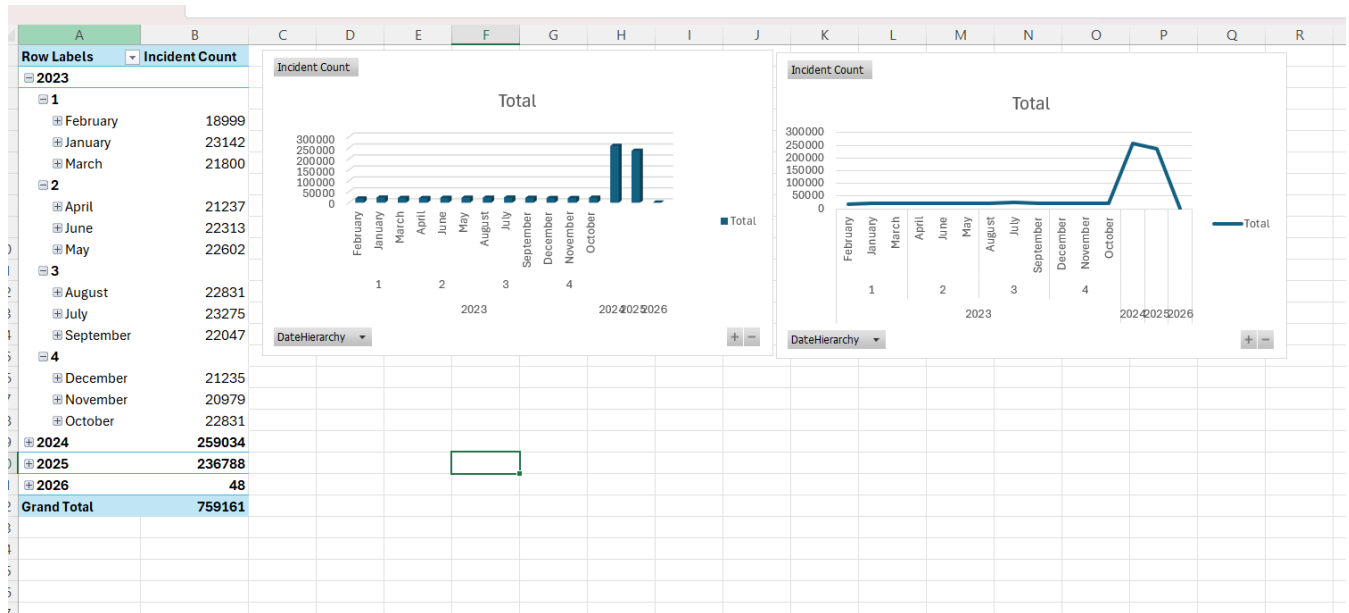
- A **Column or Bar Chart** was used to show filtered results for slice and dice.
- A **PivotChart** was used to show the changed orientation during the pivot operation.

These charts made the OLAP operations easier to understand visually.

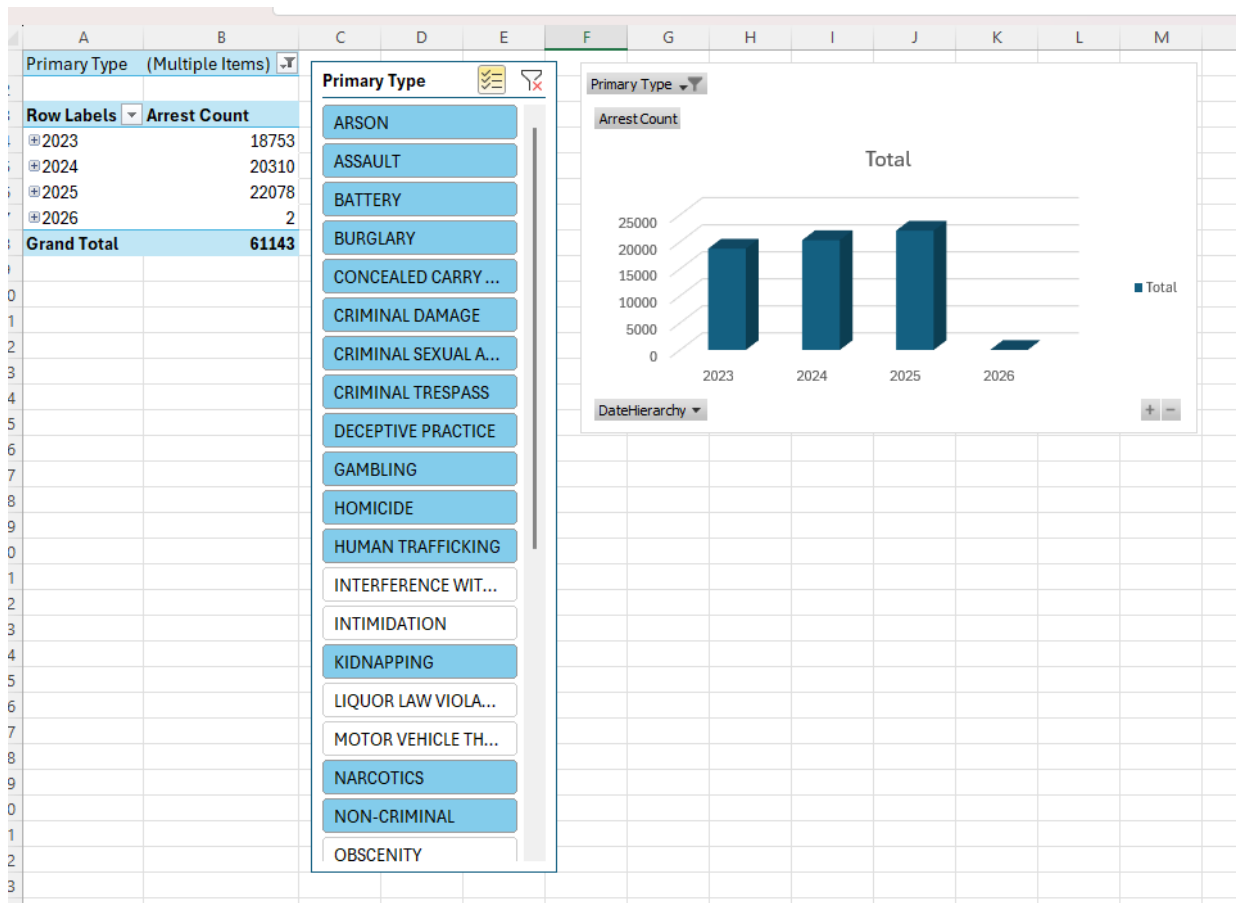
Rollup OLAP Operation



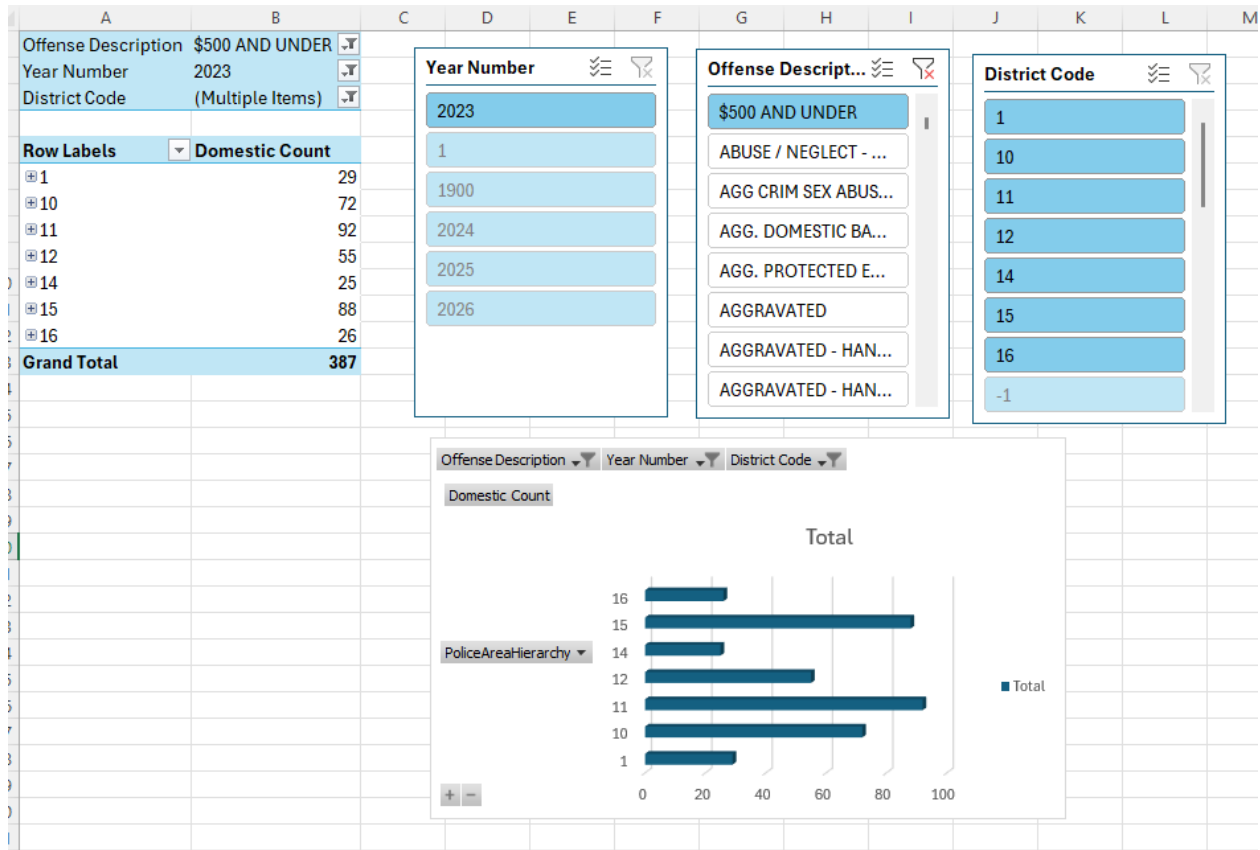
DrillDown OLAP Operation



Slice OLAP Operation

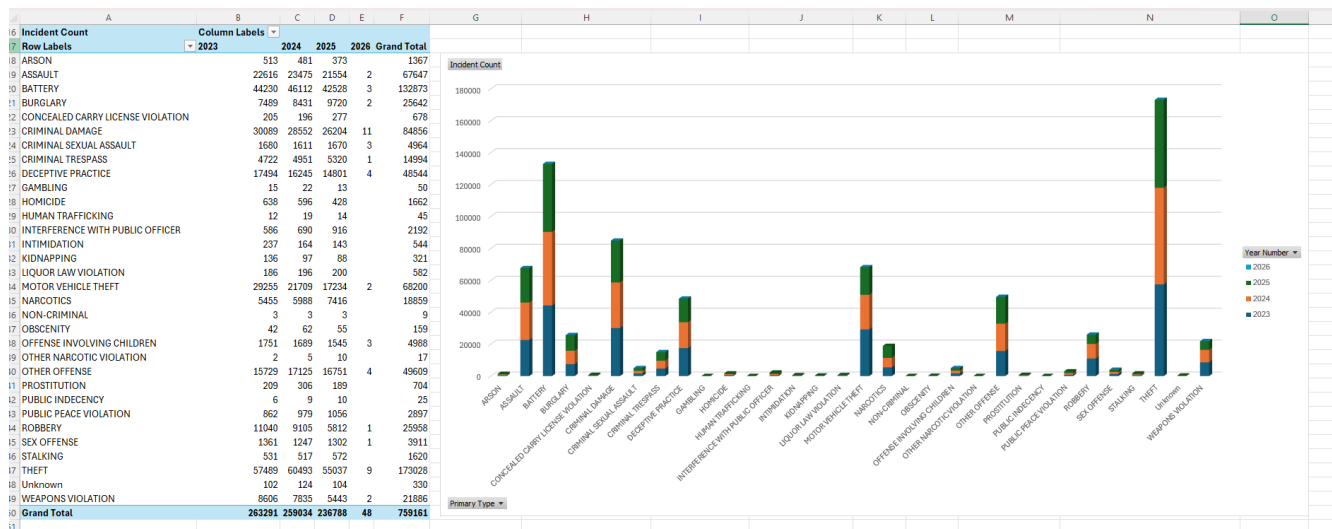


Dice OLAP Operation

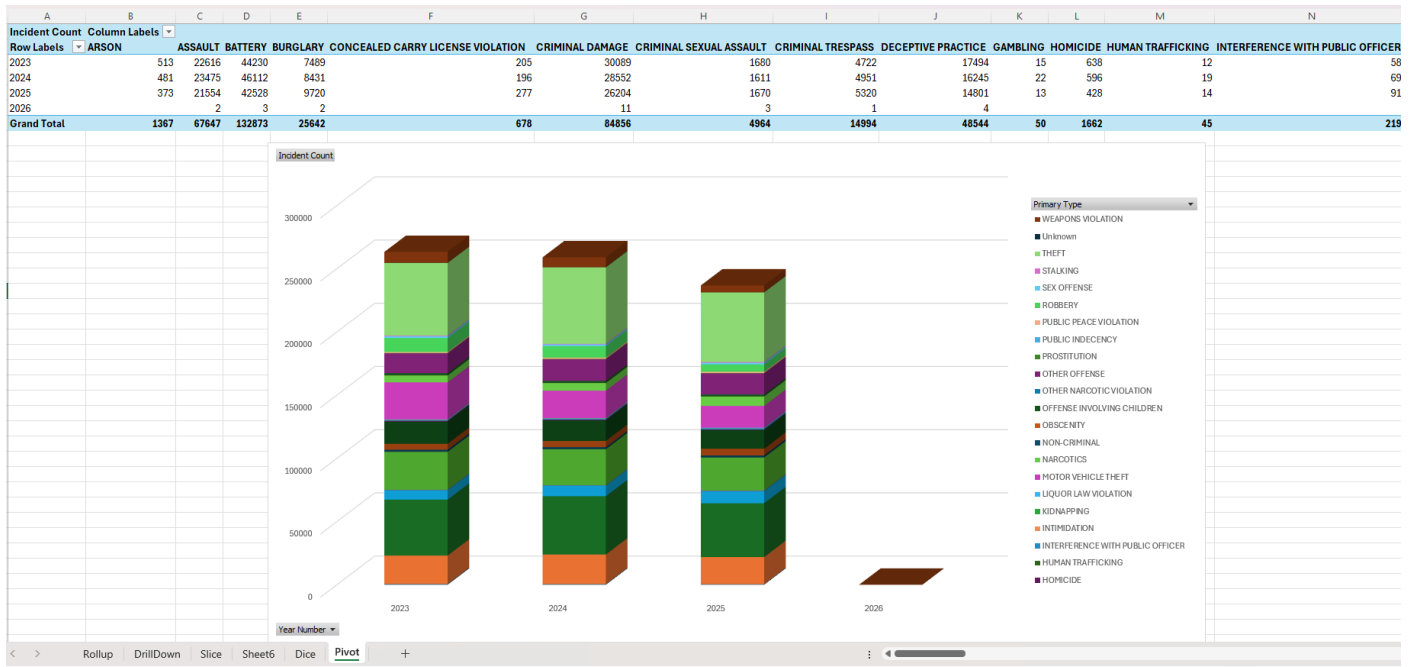


Pivot OLAP Operation

Initial Phase



Layout after Change



Step 4: PowerBI Reports

4.1 Introduction

In this step, Power BI Desktop was used to create four interactive reports based on the DWBI_Warehouse data warehouse. The reports were designed to demonstrate matrix reporting, cascading slicers, drill-down analysis, and drill-through navigation. The required tables were loaded into Power BI, relationships were configured, and visuals were created.

4.2 Data Preparation

The following tables were imported from the DWBI_Warehouse database into Power BI Desktop:

- FactCrimeIncident
- DimDate
- DimOffense
- DimLocation
- DimPoliceArea

These tables were sufficient to support the required Power BI reports.

DateKey	FullDate	DayNumberOfWeek	DayName	DayNumberOfMonth	MonthNumber	MonthName	QuarterNumber	YearNumber	WeekendFlag
20240815	15 August 2024	5	Thursday	15	8	August	3	2024	N
20240816	16 August 2024	6	Friday	16	8	August	3	2024	N
20240819	19 August 2024	2	Monday	19	8	August	3	2024	N
20240820	20 August 2024	3	Tuesday	20	8	August	3	2024	N
20240821	21 August 2024	4	Wednesday	21	8	August	3	2024	N
20240822	22 August 2024	5	Thursday	22	8	August	3	2024	N
20240823	23 August 2024	6	Friday	23	8	August	3	2024	N
20240826	26 August 2024	2	Monday	26	8	August	3	2024	N
20240827	27 August 2024	3	Tuesday	27	8	August	3	2024	N
20240828	28 August 2024	4	Wednesday	28	8	August	3	2024	N
20240829	29 August 2024	5	Thursday	29	8	August	3	2024	N
20240830	30 August 2024	6	Friday	30	8	August	3	2024	N
20240902	02 September 2024	2	Monday	2	9	September	3	2024	N
20240903	03 September 2024	3	Tuesday	3	9	September	3	2024	N
20240904	04 September 2024	4	Wednesday	4	9	September	3	2024	N
20240905	05 September 2024	5	Thursday	5	9	September	3	2024	N
20240906	06 September 2024	6	Friday	6	9	September	3	2024	N
20240909	09 September 2024	2	Monday	9	9	September	3	2024	N
20240910	10 September 2024	3	Tuesday	10	9	September	3	2024	N
20240911	11 September 2024	4	Wednesday	11	9	September	3	2024	N
20240912	12 September 2024	5	Thursday	12	9	September	3	2024	N
20240913	13 September 2024	6	Friday	13	9	September	3	2024	N
20240916	16 September 2024	2	Monday	16	9	September	3	2024	N
20240917	17 September 2024	3	Tuesday	17	9	September	3	2024	N
20240918	18 September 2024	4	Wednesday	18	9	September	3	2024	N
20240919	19 September 2024	5	Thursday	19	9	September	3	2024	N
20240920	20 September 2024	6	Friday	20	9	September	3	2024	N
20240923	23 September 2024	2	Monday	23	9	September	3	2024	N
20240924	24 September 2024	3	Tuesday	24	9	September	3	2024	N
20240925	25 September 2024	4	Wednesday	25	9	September	3	2024	N
20240926	26 September 2024	5	Thursday	26	9	September	3	2024	N
20240927	27 September 2024	6	Friday	27	9	September	3	2024	N
20240930	30 September 2024	2	Monday	30	9	September	3	2024	N
20240101	01 January 2024	2	Monday	1	1	January	1	2024	N
20240102	02 January 2024	3	Tuesday	2	1	January	1	2024	N
20240103	03 January 2024	4	Wednesday	3	1	January	1	2024	N
20240104	04 January 2024	5	Thursday	4	1	January	1	2024	N

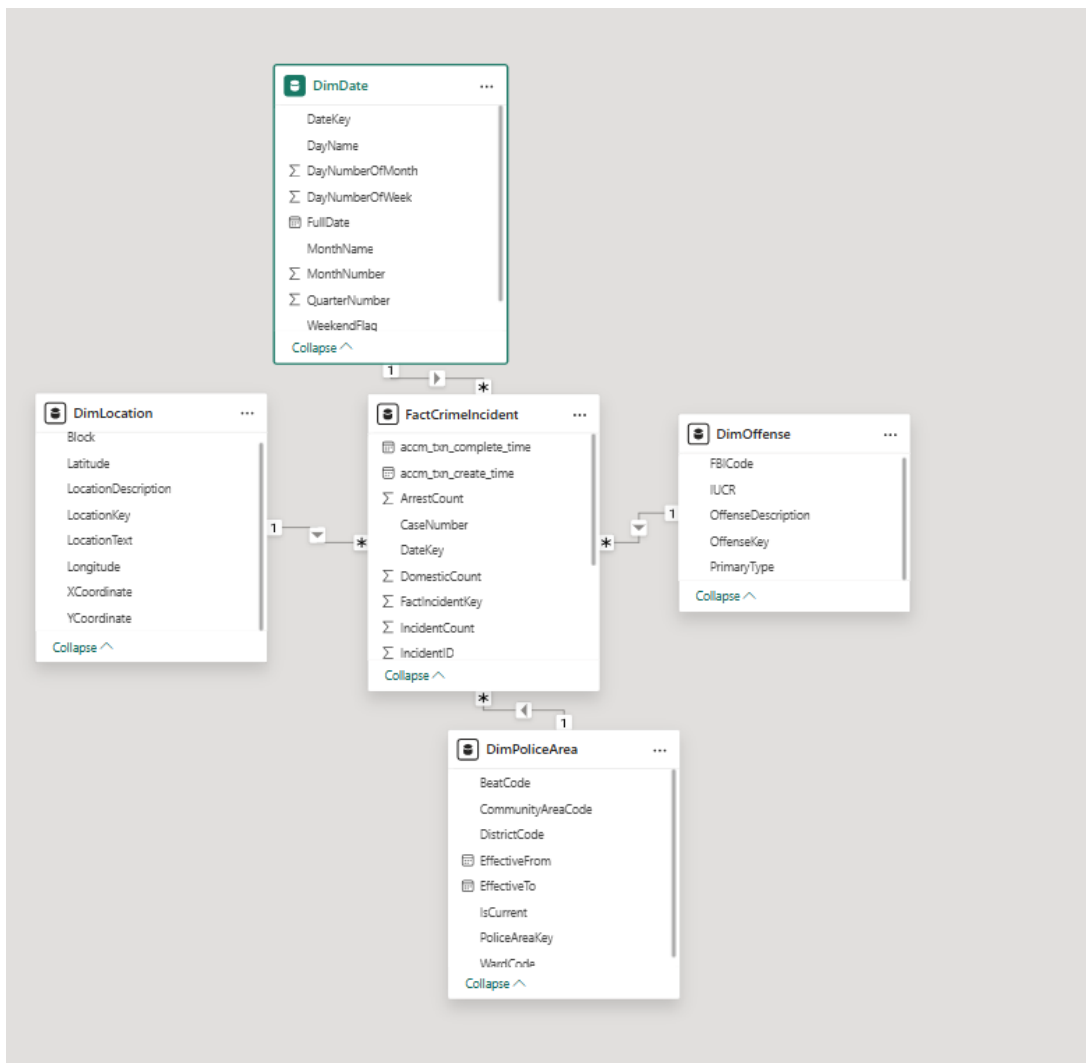
4.3 Data Modeling

After loading the tables, the model relationships were checked in **Model view**. A star-schema structure was used with **FactCrimeIncident** as the central fact table and the remaining tables as dimensions.

The relationships used were:

- `FactCrimeIncident[DateKey] → DimDate[DateKey]`
- `FactCrimeIncident[OffenseKey] → DimOffense[OffenseKey]`
- `FactCrimeIncident[LocationKey] → DimLocation[LocationKey]`
- `FactCrimeIncident[PoliceAreaKey] → DimPoliceArea[PoliceAreaKey]` →

This model ensured that the visuals aggregated and filtered correctly.



4.4 Measures Used

The reports used the following measures from the fact table:

- IncidentCount
- ArrestCount
- DomesticCount

In the visuals, these appeared as aggregated values such as:

- Sum of IncidentCount
- Sum of ArrestCount
- Sum of DomesticCount

These measures were used throughout all report pages.

4.5 Report 1: Matrix View of Crime Data

A matrix report was created to display tabular data with both row and column groupings.

Visual used

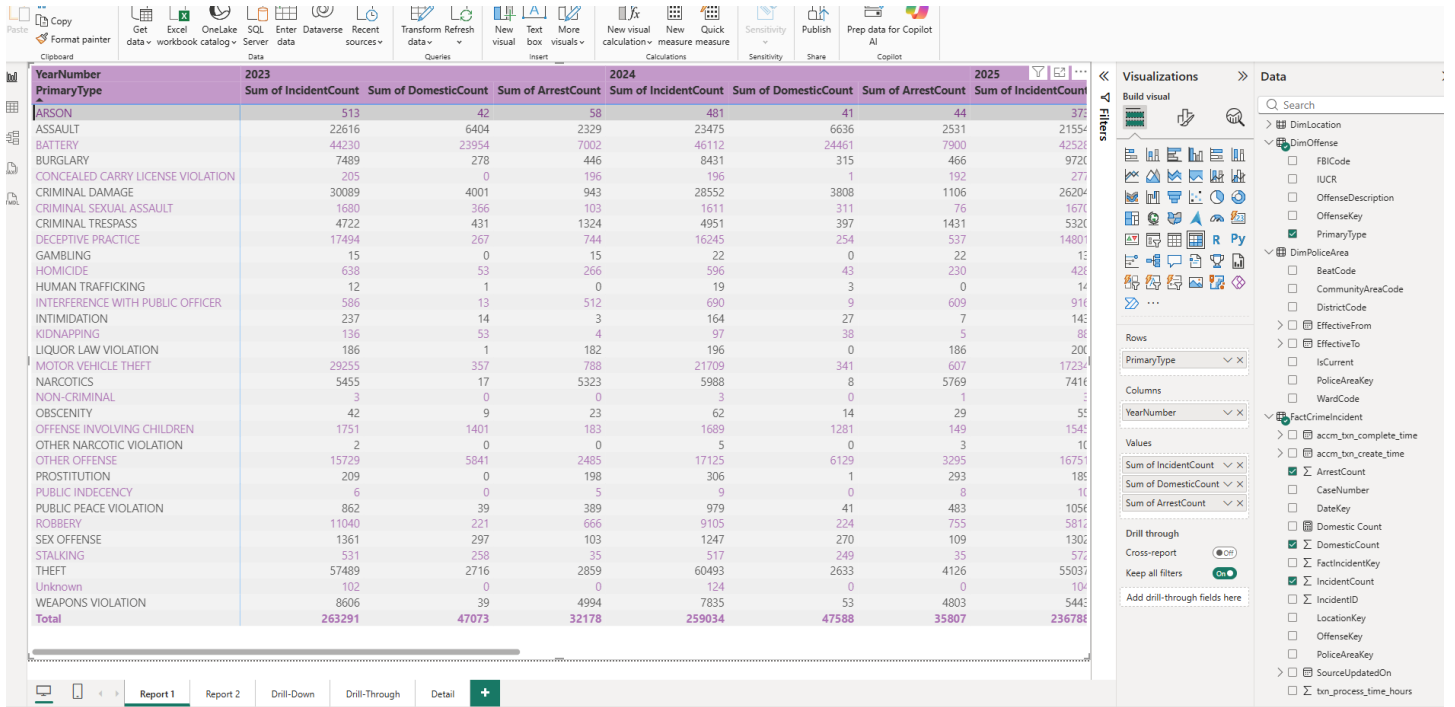
- Matrix

Fields used

- **Rows:** PrimaryType
- **Columns:** YearNumber
- **Values:**
 - Sum of IncidentCount
 - Sum of DomesticCount
 - Sum of ArrestCount

Description

This report presents crime measures by offense type across multiple years in a matrix format. The row grouping is based on offense categories, while the column grouping is based on year. This allows a structured comparison of incident, domestic, and arrest counts over time.



YearNumber	2025	2026	Total		
PrimaryType	Sum of ArrestCount	Sum of IncidentCount	Sum of DomesticCount	Sum of ArrestCount	Sum of IncidentCount
ARSON	43			1367	113
ASSAULT	2678	2	1	67647	19208
BATTERY	8122	3	3	132873	71013
BURGLARY	462	2	0	25642	902
CONCEALED CARRY LICENSE VIOLATION	270			678	1
CRIMINAL DAMAGE	1131	11	0	84856	11457
CRIMINAL SEXUAL ASSAULT	49	3	0	4964	1019
CRIMINAL TRESPASS	1679	1	0	14994	1232
DECEPTIVE PRACTICE	448	4	0	48544	766
GAMBLING	12			50	0
HOMICIDE	142			1662	146
HUMAN TRAFFICKING	1			45	6
INTERFERENCE WITH PUBLIC OFFICER	838			2192	43
INTIMIDATION	3			544	65
KIDNAPPING	7			321	123
LIQUOR LAW VIOLATION	186			582	1
MOTOR VEHICLE THEFT	580	2	0	68200	1065
NARCOTICS	7031			18859	36
NON-CRIMINAL	3			9	0
OBSCENITY	28			159	39
OFFENSE INVOLVING CHILDREN	111	3	3	4988	3849
OTHER NARCOTIC VIOLATION	5			17	0
OTHER OFFENSE	3393	4	2	49609	17950
PROSTITUTION	172			704	1
PUBLIC INDECENCY	10			25	0
PUBLIC PEACE VIOLATION	535			2897	113
ROBBERY	553	1	0	25958	654
SEX OFFENSE	109	1	0	3911	870
STALKING	25			1620	780
THEFT	4954	9	2	173028	8171
Unknown	0			330	0
WEAPONS VIOLATION	4245	2	0	21886	156
Total	37825	48	11	759161	139779

Figure 4: Matrix report showing crime measures by offense type and year

4.6 Report 2: Cascading Slicers and Visual Insights

A report page was created using multiple slicers and multiple visuals to demonstrate interactive filtering.

Slicers used

- DistrictCode
- BeatCode

Visuals used

- **Column chart:** Sum of IncidentCount by PrimaryType
- **Bar chart:** Sum of DomesticCount by DistrictCode
- **Card:** Sum of ArrestCount

Description

This report demonstrates interactive filtering using multiple slicers. The DistrictCode slicer was used as the first filter and the BeatCode slicer as the second filter. Selecting a district dynamically narrowed the beat options, demonstrating cascading filtering behavior. The charts and card updated according to slicer selections and provided insights into the filtered data.

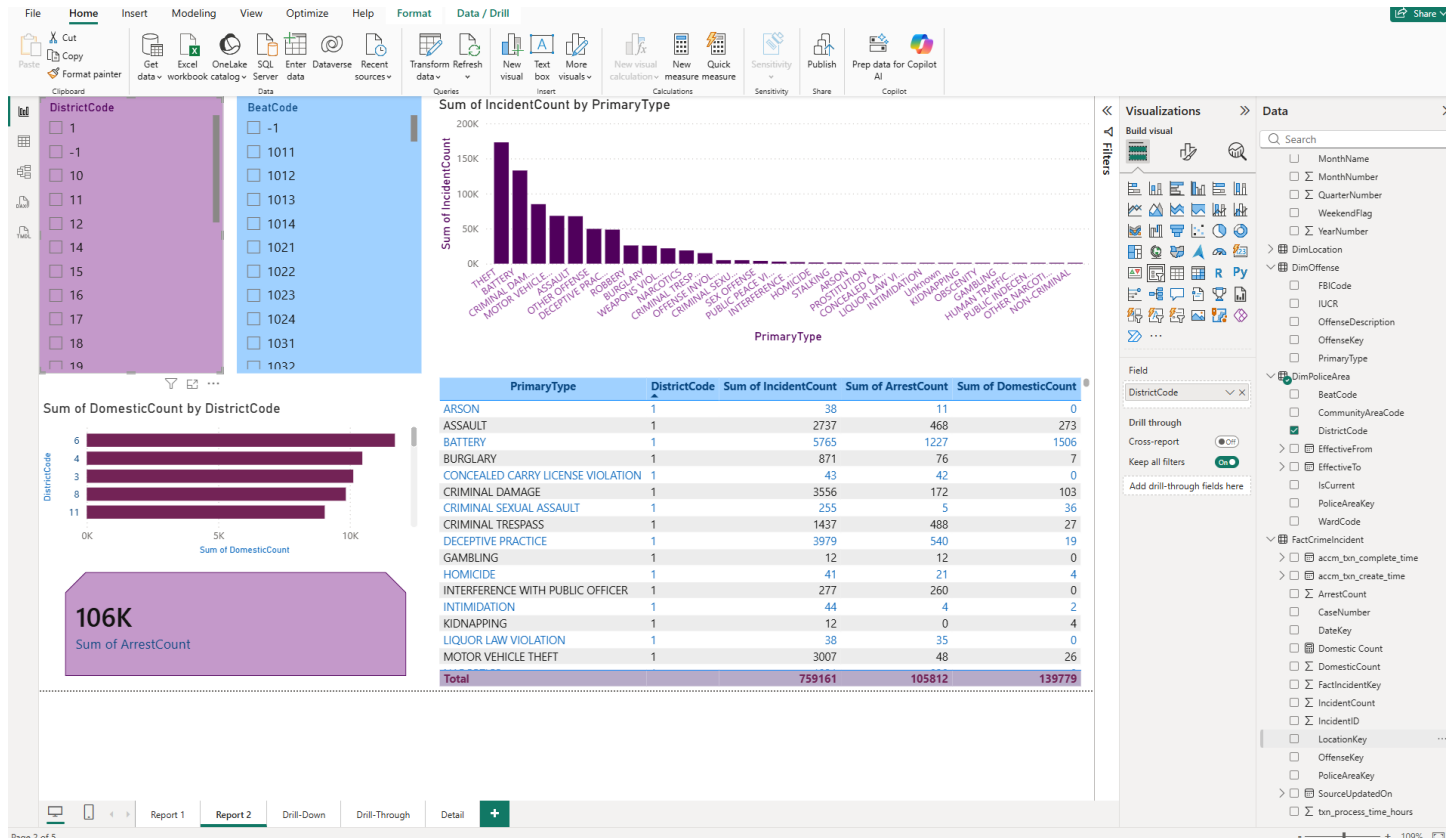


Figure 5: Report 2 demonstrating cascading slicers and interactive visual filtering

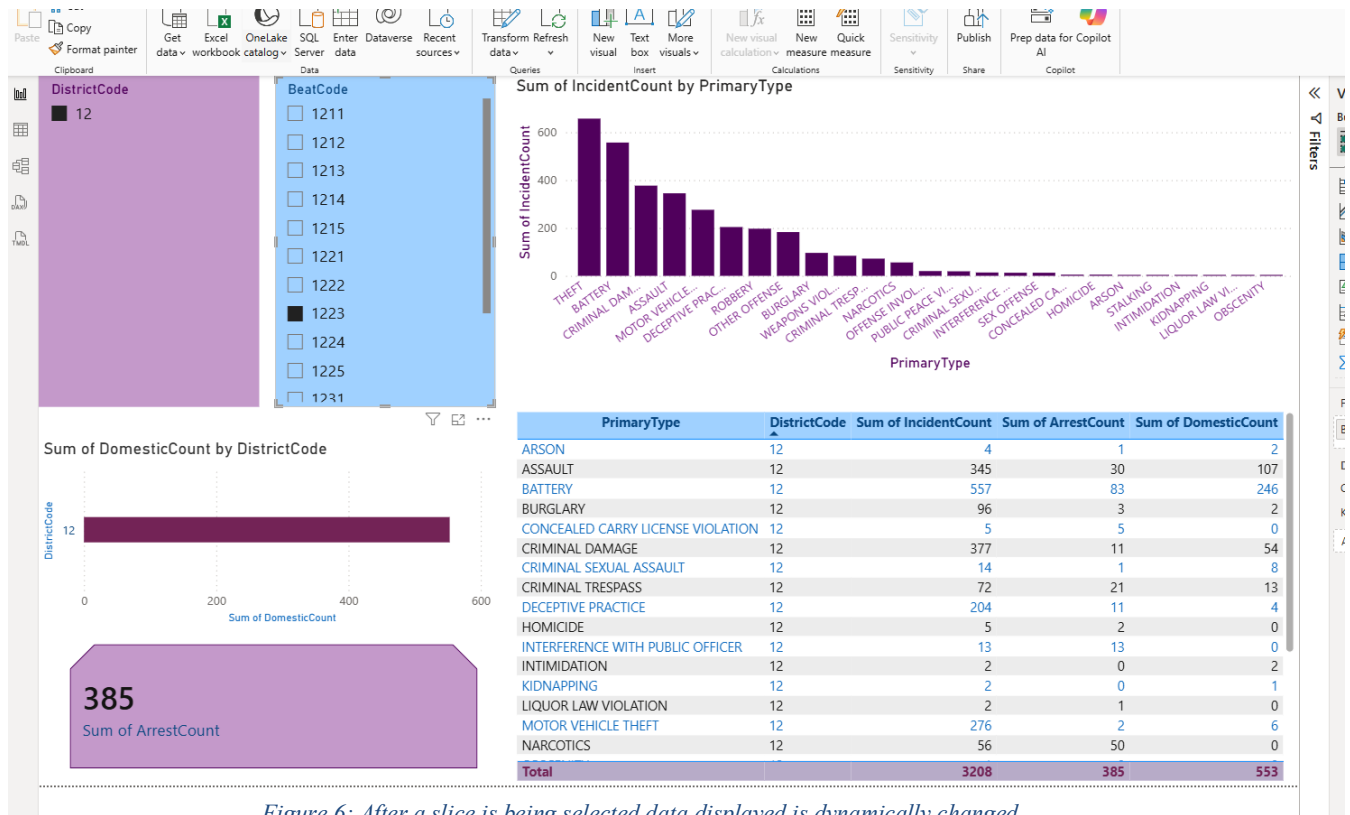


Figure 6: After a slice is being selected data displayed is dynamically changed

4.7 Report 3: Drill-Down Analysis by Date Hierarchy

A drill-down report was created to allow hierarchical exploration of crime data by time.

Visual used

- Clustered column chart

Fields used

- **X-axis:** date hierarchy (YearNumber → QuarterNumber → MonthName)
- **Y-axis:** Sum of IncidentCount

Description

This report was designed to support drill-down analysis across the date hierarchy. The chart initially displayed data at the year level. Using the drill controls, the report was expanded to lower levels such as quarter and month. This enabled users to move from summarized time-based analysis to more detailed views.

Drill-down report using the date hierarchy

Chart at year level

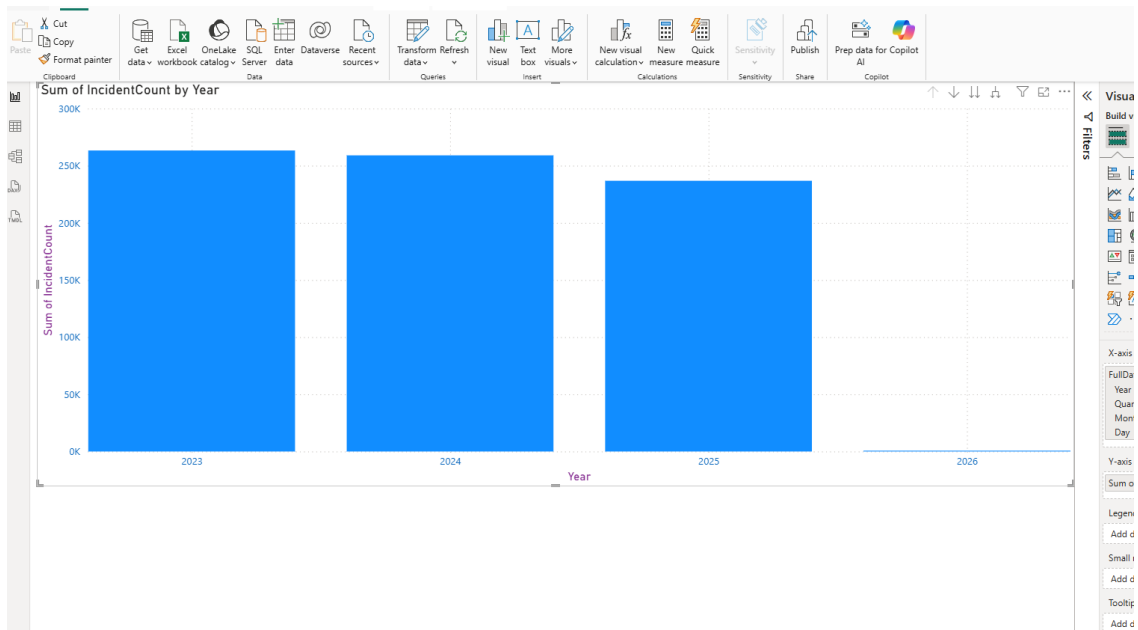
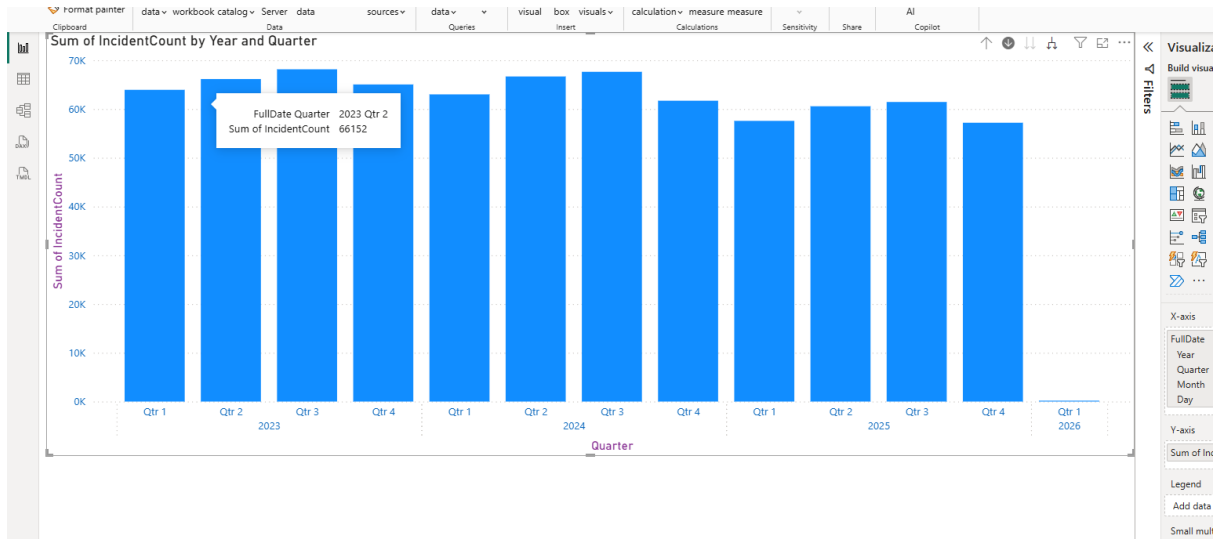
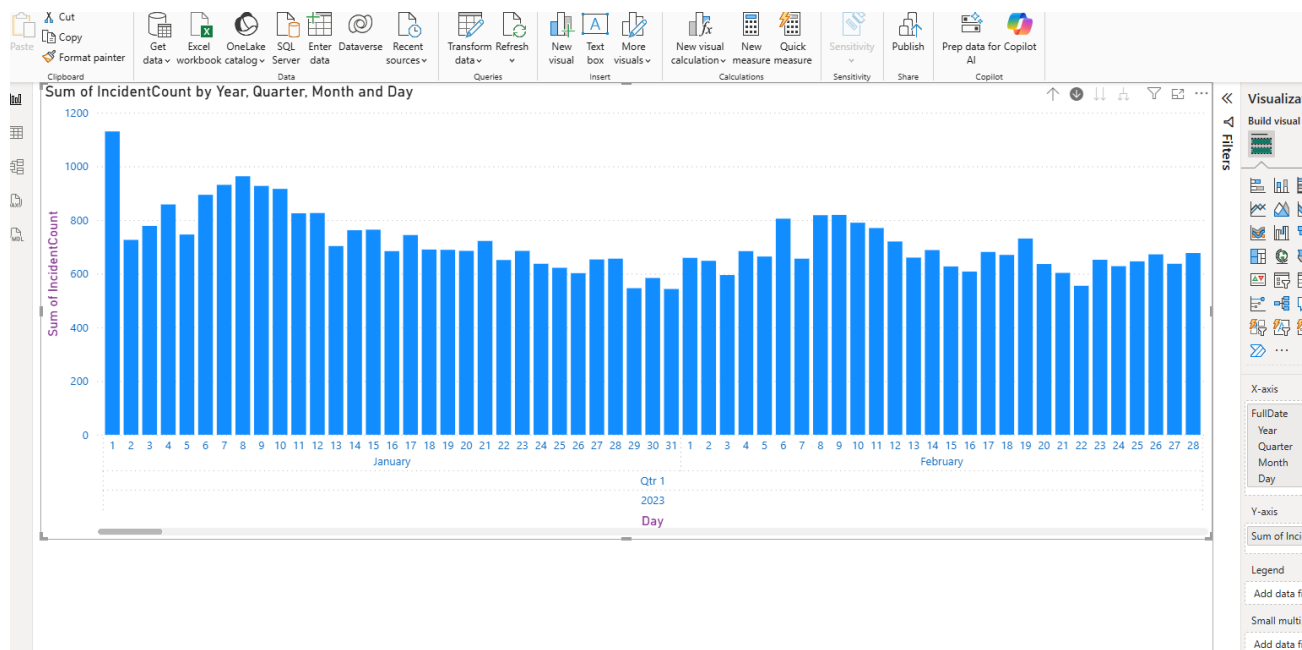
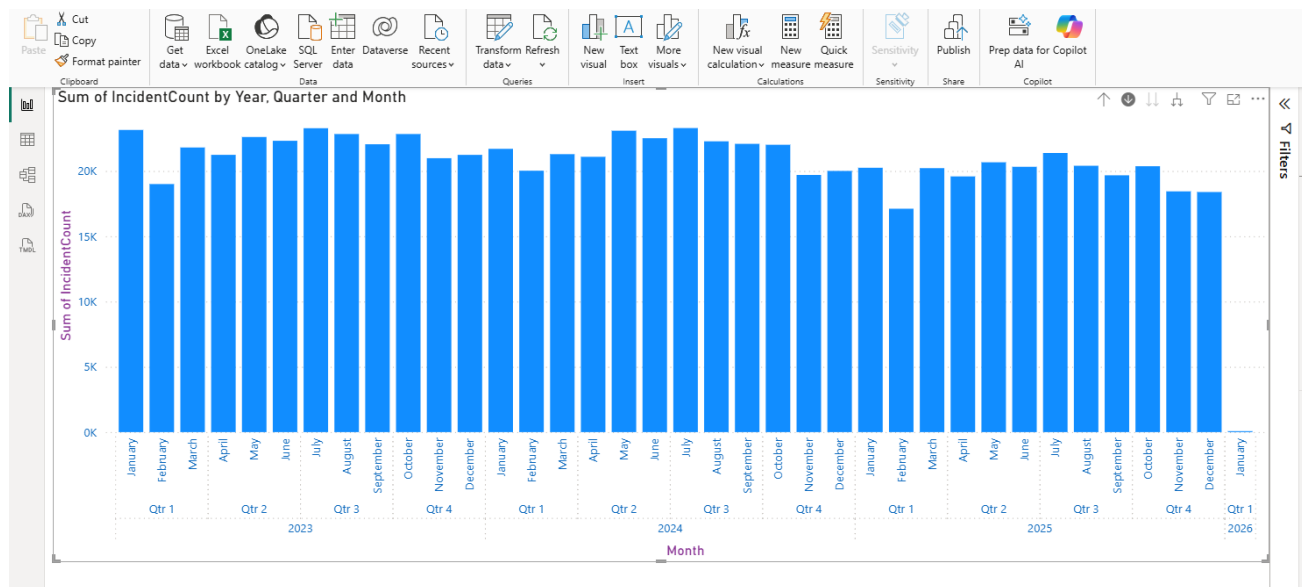


Chart after drilling down to quarter / month levels





4.8 Report 4: Drill-Through Navigation

A drill-through report was created using a summary page and a filtered detail page.

Summary page visual

- **Bar chart**
- **Axis:** `PrimaryType`
- **Values:** `Sum of IncidentCount`

Detail page configuration

- **Drill-through field:** `PrimaryType`
- **Detail visual:** table or matrix

Fields shown in the detail visual

- `PrimaryType`
- `YearNumber`
- `DistrictCode`
- `IncidentCount`
- `ArrestCount`
- `DomesticCount`

Additional design element

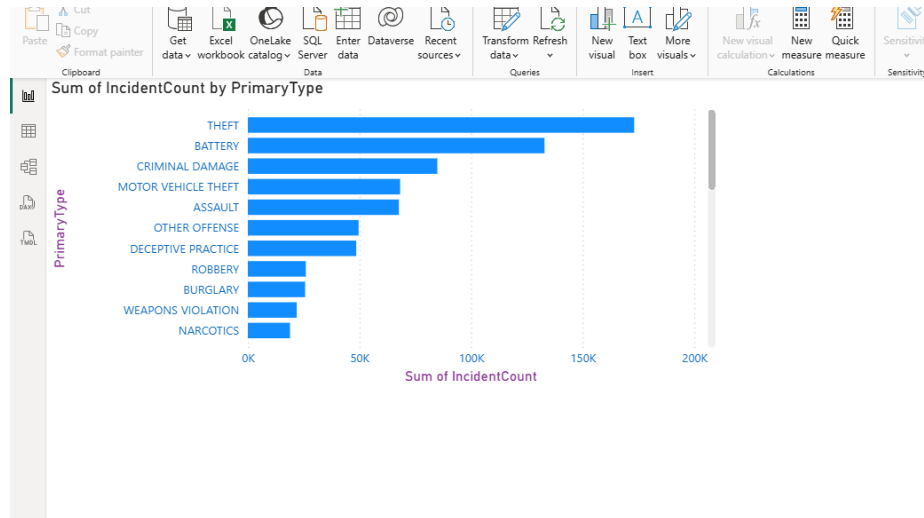
- **Back button** on the detail page

Description

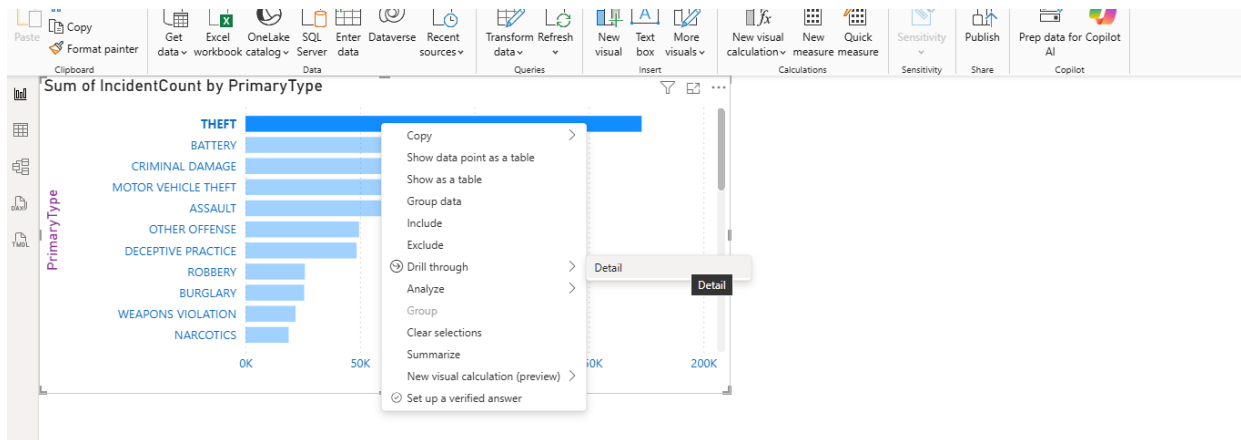
This report demonstrates drill-through functionality in Power BI. The summary page allows users to right-click an offense category in the chart and navigate to a detail page. The detail page then shows data filtered to the selected offense category, allowing more detailed analysis without losing navigation context.

Drill-through report showing navigation from summary view to filtered detail view

Summary page



Right-click menu showing drill-through option



Detail page after drill-through navigation



PrimaryType	Sum of YearNumber	DistrictCode	Sum of IncidentCount	Sum of ArrestCount	Sum of DomesticCount
THEFT	2222231	1	15057	1892	208
THEFT	2222231	10	4640	288	382
THEFT	2222231	11	4613	104	572
THEFT	2222231	12	13213	506	309
THEFT	2222231	14	9301	948	131
THEFT	2222231	15	3501	87	513
THEFT	2222231	16	6448	269	141
THEFT	2222231	17	6723	656	116
THEFT	2222231	18	16050	1405	184
THEFT	2222231	19	15356	1002	116
THEFT	2222231	2	8943	264	535
THEFT	2222231	20	4793	326	96
THEFT	2222231	22	4580	257	339
THEFT	2222231	24	6864	646	202
THEFT	2222231	25	8054	1013	372
THEFT	2222231	3	6378	219	660
THEFT	2222231	31	6	1	0
THEFT	2222231	4	6678	185	662
THEFT	2222231	5	4787	478	603
THEFT	2222231	6	7052	352	797
THEFT	2222231	7	4134	90	559
THEFT	2222231	8	9380	498	433
THEFT	2222231	9	6477	453	241
Total	2222231		173028	11939	8171



Report 1Report 2Drill-DownDrill-ThroughDetail

4.9 Result Comparison Between SSMS and Power BI

To confirm the correctness of the Power BI results, selected values from the report visuals were compared against the corresponding results in SQL Server Management Studio (SSMS). Matching totals between SSMS query outputs and Power BI visual outputs confirmed that the imported data, model relationships, and aggregations were functioning correctly.

SSMS query result

```
50
51 USE DWBI_Warehouse;
52
53 SELECT
54     d.YearNumber,
55     o.PrimaryType,
56     SUM(f.IncidentCount) AS IncidentCount,
57     SUM(f.ArrestCount) AS ArrestCount,
58     SUM(f.DomesticCount) AS DomesticCount
59 FROM FactCrimeIncident f
60 JOIN DimDate d
61     ON f.DateKey = d.DateKey
62 JOIN DimOffense o
63     ON f.OffenseKey = o.OffenseKey
64 GROUP BY d.YearNumber, o.PrimaryType
65 ORDER BY d.YearNumber, o.PrimaryType;
```

100 % 3 0

Results Messages

	YearNumber	PrimaryType	IncidentCount	ArrestCount	DomesticCount
1	2023	ARSON	513	58	42
2	2023	ASSAULT	22616	2329	6404
3	2023	BATTERY	44230	7002	23954
4	2023	BURGLARY	7489	446	278
5	2023	CONCEALED CARRY LICENSE VIOLATION	205	196	0
6	2023	CRIMINAL DAMAGE	30089	943	4001
7	2023	CRIMINAL SEXUAL ASSAULT	1680	103	366
8	2023	CRIMINAL TRESPASS	4722	1324	431
9	2023	DECEPTIVE PRACTICE	17494	744	267
10	2023	GAMBLING	15	15	0
11	2023	HOMICIDE	638	266	53
12	2023	HUMAN TRAFFICKING	12	0	1
13	2023	INTERFERENCE WITH PUBLIC OFFICER	586	512	13
14	2023	INTIMIDATION	237	3	14
15	2023	KIDNAPPING	136	4	53
16	2023	LIQUOR LAW VIOLATION	186	182	1
17	2023	MOTOR VEHICLE THEFT	29255	788	357
18	2023	NARCOTICS	5455	5323	17
19	2023	NON-CRIMINAL	3	0	0

Power BI visual showing the same corresponding value

YearNumber	2023		202
PrimaryType	Sum of IncidentCount	Sum of DomesticCount	Sum of ArrestCount
ARSON	513	42	58
ASSAULT	22616	6404	2329
BATTERY	44230	23954	7002
BURGLARY	7489	278	446
CONCEALED CARRY LICENSE VIOLATION	205	0	196
CRIMINAL DAMAGE	30089	4001	943
CRIMINAL SEXUAL ASSAULT	1680	366	103
CRIMINAL TRESPASS	4722	431	1324
DECEPTIVE PRACTICE	17494	267	744
GAMBLING	15	0	15
HOMICIDE	638	53	266
HUMAN TRAFFICKING	12	1	0
INTERFERENCE WITH PUBLIC OFFICER	586	13	512
INTIMIDATION	237	14	3
KIDNAPPING	136	53	4
LIQUOR LAW VIOLATION	186	1	182
MOTOR VEHICLE THEFT	29255	357	788
NARCOTICS	5455	17	5323
NON-CRIMINAL	3	0	0
OBSCENITY	42	9	23
OFFENSE INVOLVING CHILDREN	1751	1401	183
OTHER NARCOTIC VIOLATION	2	0	0
OTHER OFFENSE	15729	5841	2485
PROSTITUTION	209	0	198
PUBLIC INDECENCY	6	0	5
PUBLIC PEACE VIOLATION	862	39	389
ROBBERY	11040	221	666
SEX OFFENSE	1361	297	103
STALKING	531	258	35
THEFT	57489	2716	2859
Unknown	102	0	0
WEAPONS VIOLATION	8606	39	4994
Total	263291	47073	32178

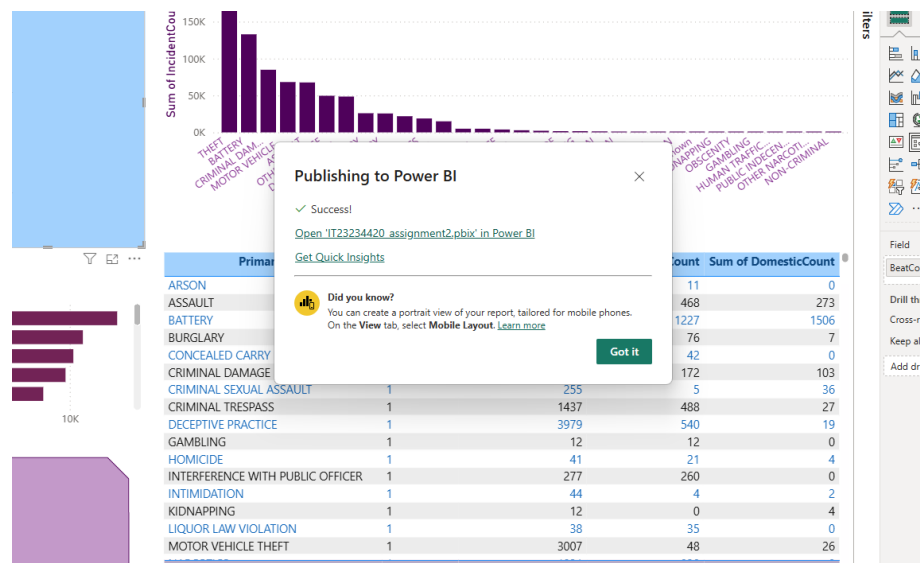
4.10 Publishing to Power BI Service

After completing the report pages, the Power BI file was published to **Power BI Service**. The report was then opened online to demonstrate the final output in the Power BI Service environment.

[View PowerBI Dashboard Here](#)

Published Power BI report opened in Power BI Service

Publish success message



Report opened in Power BI Service online

The screenshot shows the Power BI Service online report. The table displays crime statistics for 2023, 2024, and 2025, including the sum of incident counts, domestic counts, and arrest counts for various crime types.

YearNumber	PrimaryType	2023 Sum of IncidentCount	2023 Sum of DomesticCount	2023 Sum of ArrestCount	2024 Sum of IncidentCount	2024 Sum of DomesticCount	2024 Sum of ArrestCount	2025 Sum of IncidentCount
2023	ARSON	513	42	58	481	41	44	371
2023	ASSAULT	22616	6404	2329	23475	6636	2531	21554
2023	BATTERY	44230	23954	7002	46112	24461	7900	42528
2023	BURGLARY	7489	278	446	8431	315	466	9728
2023	CONCEALED CARRY LICENSE VIOLATION	205	0	196	196	1	192	277
2023	CRIMINAL DAMAGE	30089	4001	943	28552	3808	1106	26204
2023	CRIMINAL SEXUAL ASSAULT	1680	366	103	1611	311	76	1671
2023	CRIMINAL TRESPASS	4722	431	1324	4951	397	1431	5328
2023	DECEPTIVE PRACTICE	17494	267	744	16245	254	537	14801
2023	GAMBLING	15	0	15	22	0	22	12
2023	HOMICIDE	638	53	266	596	43	230	428
2023	HUMAN TRAFFICKING	12	1	0	19	3	0	14
2023	INTERFERENCE WITH PUBLIC OFFICER	586	13	512	690	9	609	911
2023	INTIMIDATION	237	14	3	164	27	7	141
2023	KIDNAPPING	136	53	4	97	38	5	88
2023	LIQUOR LAW VIOLATION	186	1	182	196	0	186	208
2023	MOTOR VEHICLE THEFT	29255	357	788	21709	341	607	17234
2023	NARCOTICS	5455	17	5323	5988	8	5769	7411
2023	NON-CRIMINAL	3	0	0	3	0	1	3
2023	OBSCENITY	42	9	23	62	14	29	52
2023	OFFENSE INVOLVING CHILDREN	1751	1401	183	1609	1281	149	1542
2023	OTHER NARCOTIC VIOLATION	2	0	0	5	0	3	11
2023	OTHER OFFENSE	15729	5841	2485	17125	6129	3295	16751
2023	PROSTITUTION	209	0	198	306	1	293	188
2023	PUBLIC INDECENCY	6	0	5	9	0	8	11
2023	PUBLIC PEACE VIOLATION	862	39	389	979	41	483	1054
2023	ROBBERY	11040	221	666	9105	224	755	5812
2023	SEX OFFENSE	1361	297	103	1247	270	109	1303
2023	STALKING	531	258	35	517	249	35	571
2023	THEFT	57489	2716	2859	60493	2633	4126	55031
2023	Unknown	102	0	0	124	0	0	159
2023	WEAPONS VIOLATION	8606	39	4994	7835	53	4803	5442
2023	Total	263291	47073	32178	259034	47588	35807	236781